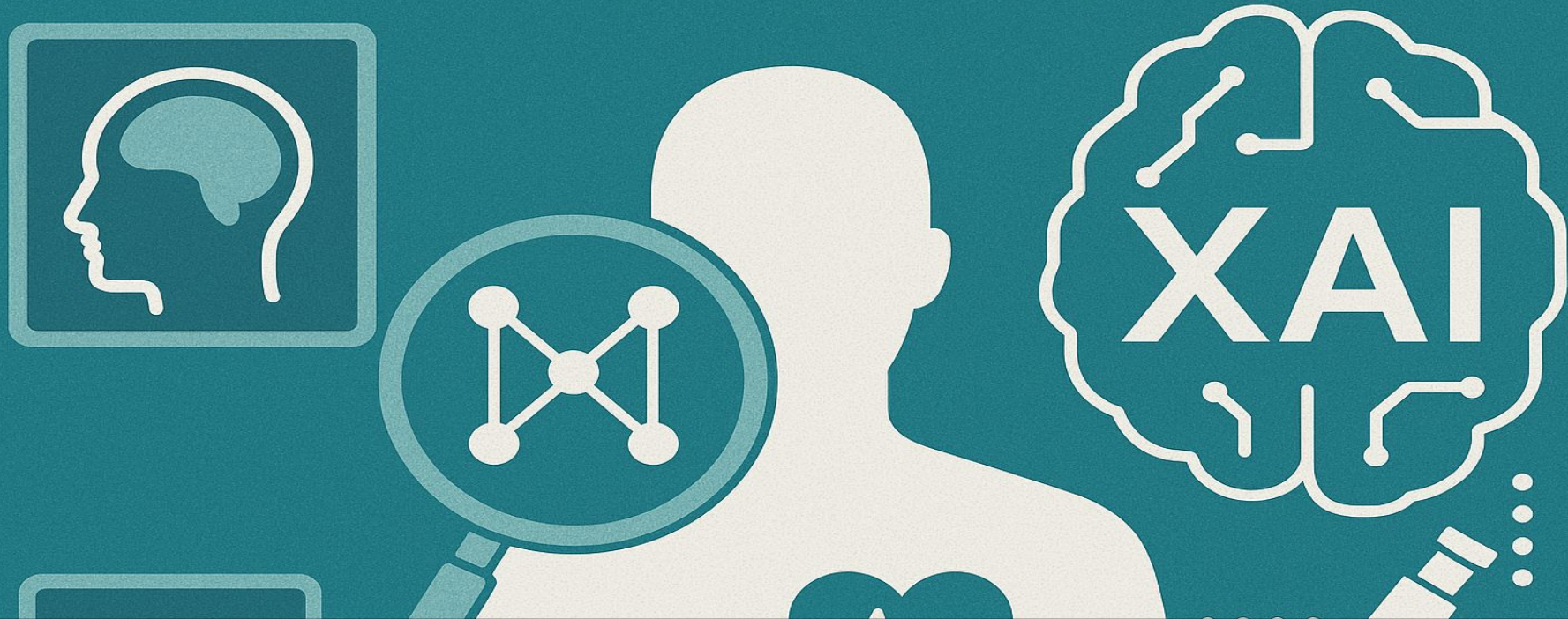




INVICTA Spring School: Explainable AI (xAI)

xAI in Healthcare: Fundamentals, Challenges, and the Road to Trustworthy AI

Wilson Silva, Utrecht University

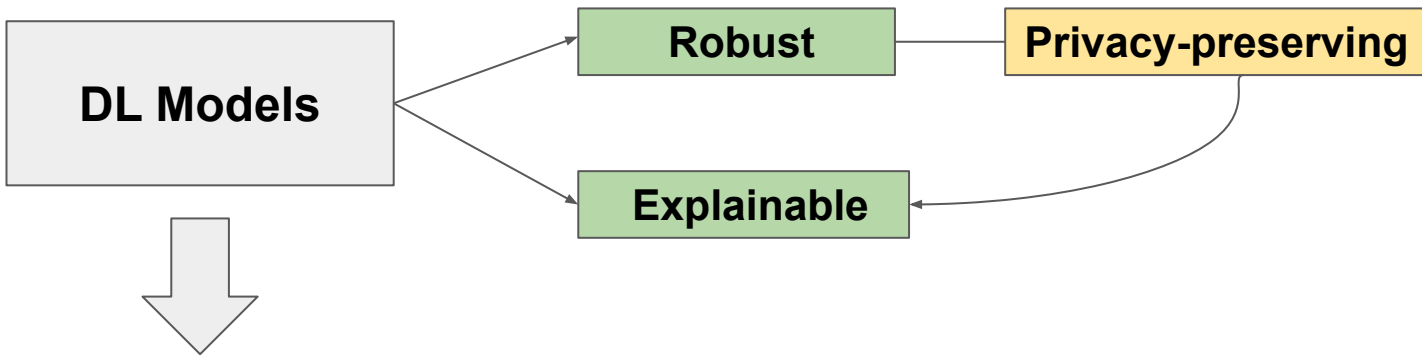
About me

- PI of the Trustworthy AI for the Life Sciences Lab
- Assistant Professor (tenure-track) at Utrecht University
 - AI Technology for Life, AI & Data Science Division
- Guest AI Researcher at the Netherlands Cancer Institute
 - Department of Radiology
- Background
 - B.Sc. + M.Sc. in ECE at the University of Porto
 - PhD in Machine Learning for Medical Image Analysis at the University of Porto
 - Visiting PhD student at University of Bern



Trustworthy AI Lab

- We develop:



Representation Learning

Trustworthy AI Lab - Postdocs and PhDs

- **Soufyan:** promoting ordinality in the latent space (omics)
- **Valentina:** robust AI for multi-center (cancer) data
- **Daan:** robust AI for immune microenvironment
- **Miriam:** CT-based (early) diagnosis using deep learning
- **Aniek:** explainable AI for multimodal life science data
- **Jan:** explaining LLMs in biology

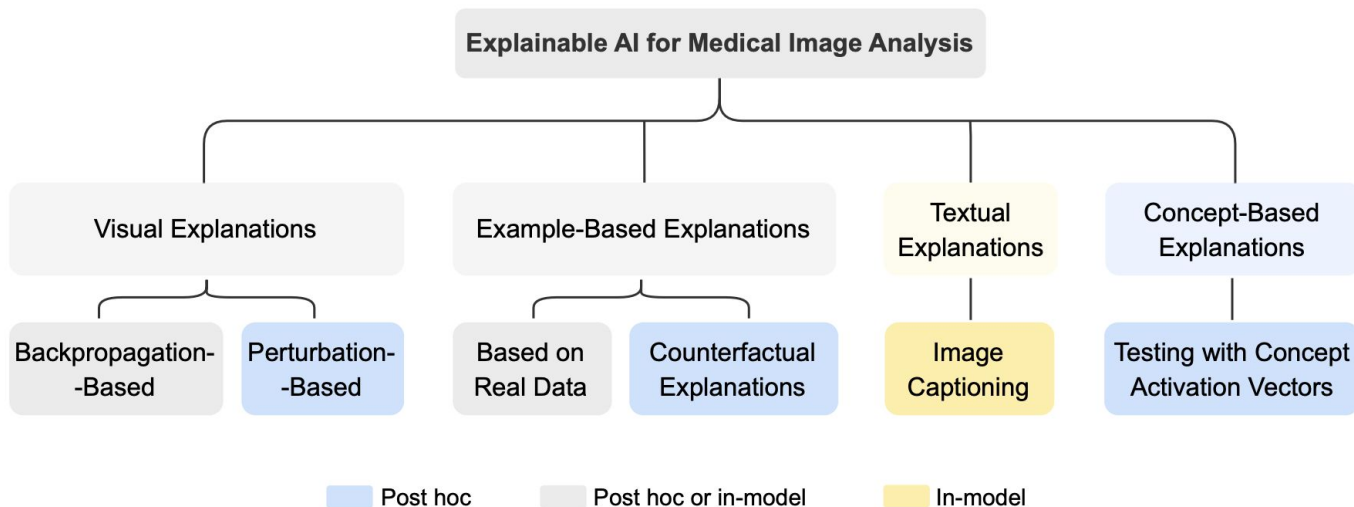


Trustworthy AI Lab - MSc students + interns

- **Carla:** robust polyp segmentation
- **Floris:** concept-based regularization
- **Cedrik:** multimodal AI for kidney cancer
- **Lena:** ordinal learning for diabetic retinopathy
- **Isabelle:** radiology image compression
- **Malaika:** federated learning
- **Inês:** model unlearning for case-based xAI
- **Gideon:** AI-based erectile function prediction
- **Sander:** BRCA mutations in pathology



Explainable ML (for Medical Image Analysis)



A Unified Approach to Interpreting Model Predictions

Scott M. Lundberg

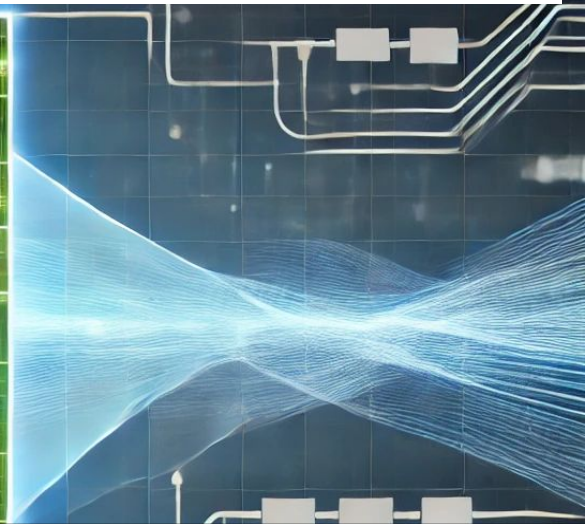
Paul G. Allen School of Computer Science
University of Washington
Seattle, WA 98105
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Su-In Lee

Paul G. Allen School of Computer Science
Department of Genome Sciences
University of Washington
Seattle, WA 98105
suinlee@cs.washington.edu

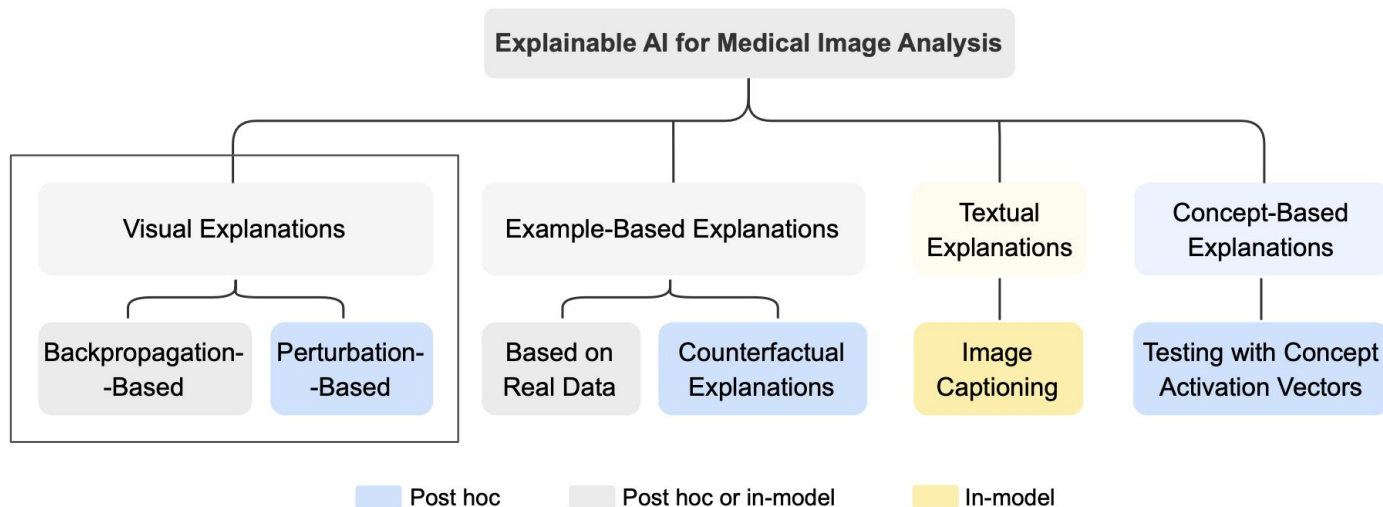
Layer-wise Relevance Propagation for Neural Networks with Local Renormalization Layers

Alexander Binder¹, Grégoire Montavon², Sebastian Bach³,
Klaus-Robert Müller^{2,4}, and Wojciech Samek³



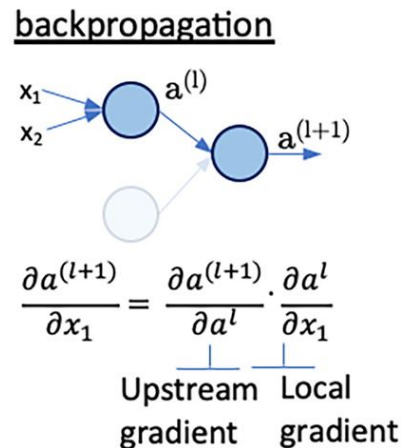
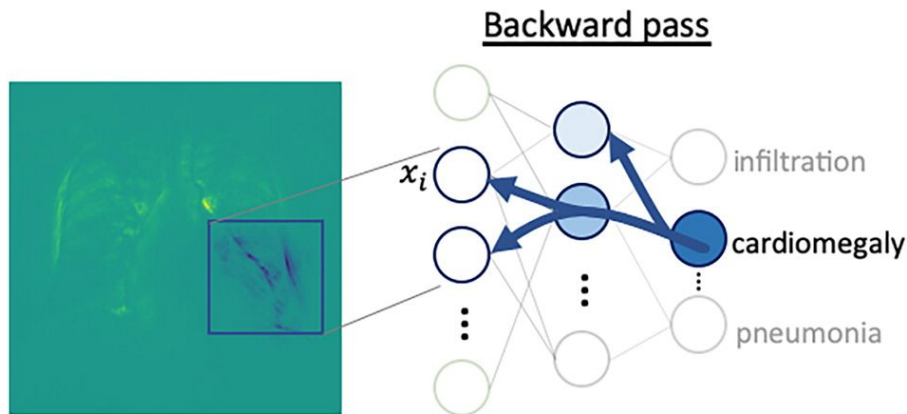
Post-hoc Feature Attribution Methods

Explainable ML (for Medical Image Analysis)



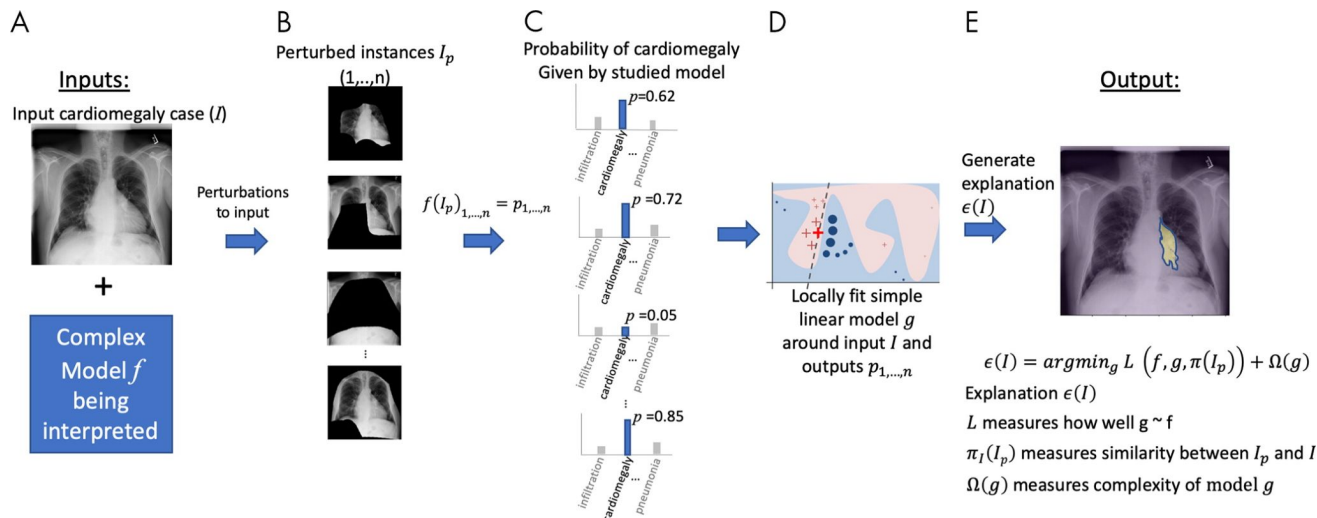
Visual Explanations

- What are **saliency maps**?
 - visual representations of **pixel contributions** to a model's decision
 - check whether **AI focus** on **pathological** areas or **confounders**



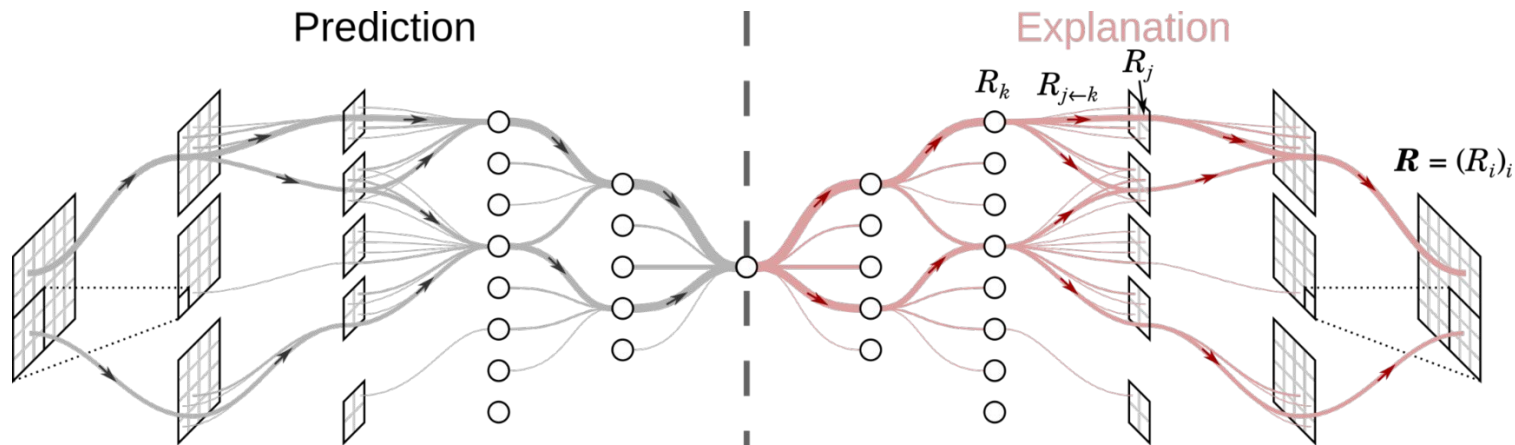
Visual Explanations

- **Saliency mapping techniques** can be divided into
 - **backpropagation** (compute pixel importance using network gradients)
 - **perturbation-based** (perturb pixels and observe changes in predictions)



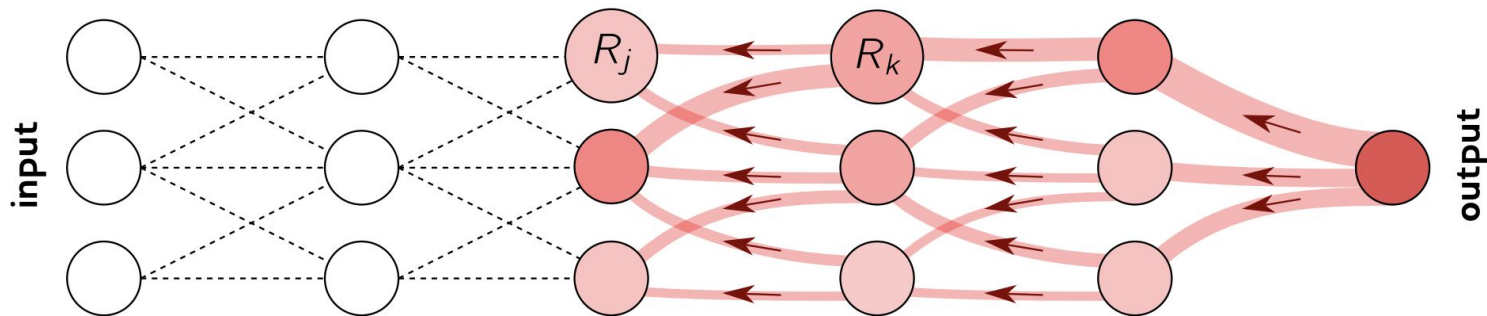
Layer-wise Relevance Propagation

- What is LRP?
 - **traces the prediction backward** from the output to the input
 - **redistributing the relevance** to highlight which input features were important



Layer-wise Relevance Propagation

- LRP procedure
 - each **neuron redistributes** to the **lower layer** as it has **received** from the **higher layer**



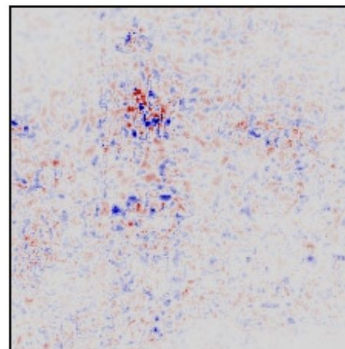
Layer-wise Relevance Propagation

- Basic Rule (LRP-0)
 - if a neuron **contributes strongly**, it gets **more relevance**
 - if a neuron **contributes weakly**, it gets **less relevance**

$$R_j = \sum_k \frac{a_j w_{jk}}{\sum_{0,j} a_j w_{jk}} R_k$$



LRP-0



Layer-wise Relevance Propagation

- Basic Rule (LRP-0)
 - Potential issues?





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$$R_j = \sum_k \frac{a_j w_{jk}}{\sum_{0,j} a_j w_{jk}} R_k$$

Layer-wise Relevance Propagation

- Basic Rule (LRP-0)
 - **small activations** and **near-zero weights** can lead to **unstable attributions**

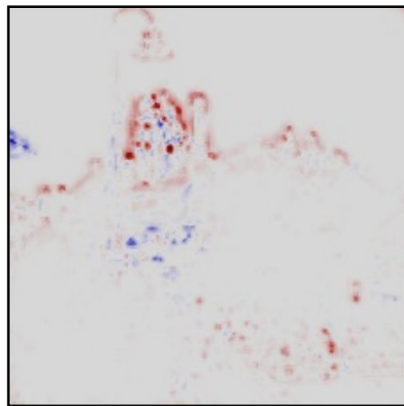
$$R_j = \sum_k \frac{a_j w_{jk}}{\sum_{0,j} a_j w_{jk}} R_k$$

Layer-wise Relevance Propagation

- Epsilon Rule (LRP- ϵ)
 - **absorb** some **relevance** when the **contributions** are **weak** or contradictory
 - as ϵ becomes **larger**, only the most salient **explanation** factors survive the absorption (**sparsity**)

$$R_j = \sum_k \frac{a_j w_{jk}}{\epsilon + \sum_{0,j} a_j w_{jk}} R_k$$

LRP- ϵ

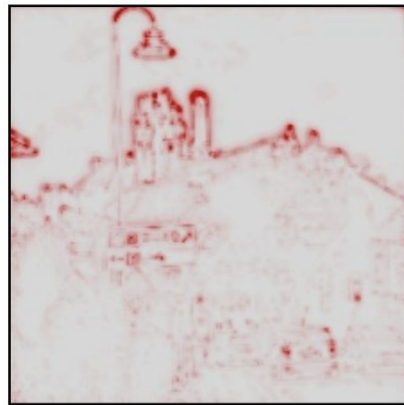


Layer-wise Relevance Propagation

- Gamma Rule (LRP- γ)
 - **favoring** the effect of **positive contributions** over negative contributions
 - as γ increases, negative contributions start to disappear

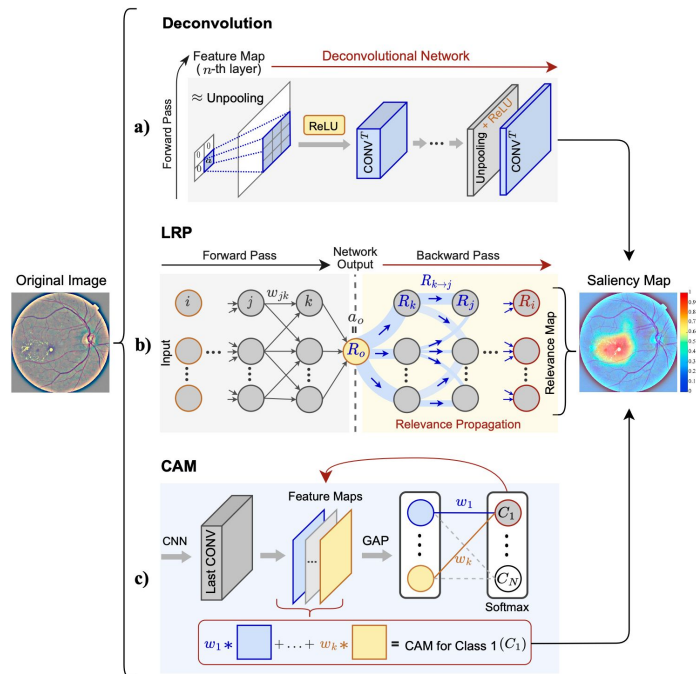
$$R_j = \sum_k \frac{a_j \cdot (w_{jk} + \gamma w_{jk}^+)}{\sum_{0,j} a_j \cdot (w_{jk} + \gamma w_{jk}^+)} R_k$$

LRP- γ



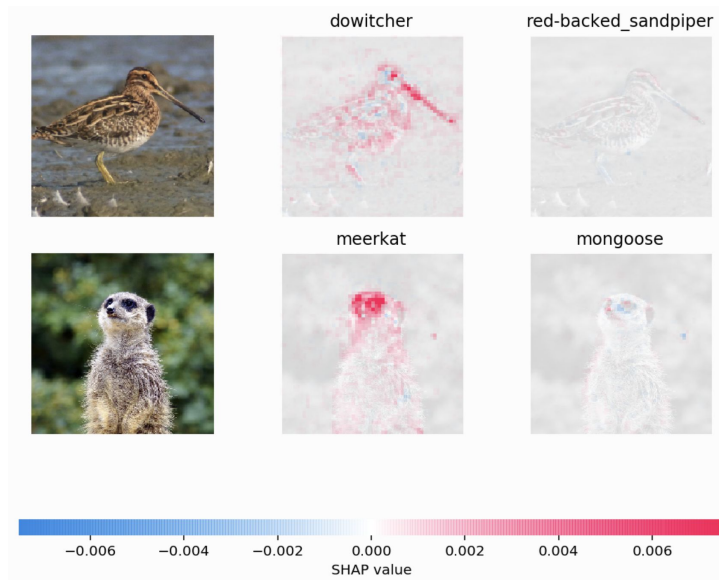
Layer-wise Relevance Propagation

- (Some) **limitations** of **LRP**
 - **depends** on **model** architecture (weights, activations)
 - **sensitive** to **rule** selection
 - **global** interpretability is **limited**
 - **theoretical guarantees** are **missing**



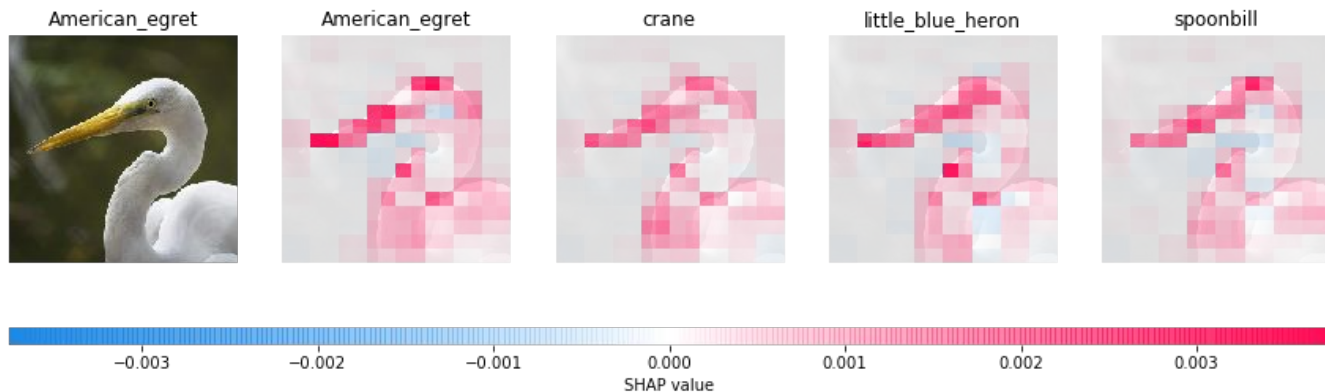
SHapley Additive exPlanations

- Transition to SHAP
 - **model-agnostic** approach to feature attribution
 - ensures **fairness** and **consistency** in **explanations**
 - **global** interpretability (limited for images though)



SHapley Additive exPlanations

- What is SHAP?
 - method that assigns an **importance score** to **each feature** based on **game theory**
 - unlike other methods, **SHAP** provides **mathematically consistent** feature **attributions**

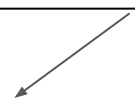


SHapley Additive exPlanations

- Shapley value definition
 - fairly distributes feature importance by considering all possible feature combinations
 - averages feature contributions across all possible coalitions

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

score with feature i in subset S



SHapley Additive exPlanations

- Shapley value definition
 - fairly distributes feature importance by considering all possible feature combinations
 - averages feature contributions across all possible coalitions

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - \underline{f(S)}]$$

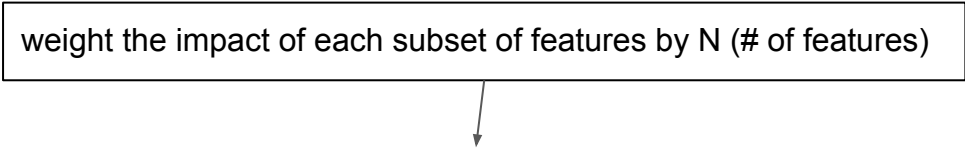
score without feature i in subset S



SHapley Additive exPlanations

- Shapley value definition
 - fairly distributes feature importance by considering all possible feature combinations
 - averages feature contributions across all possible coalitions

weight the impact of each subset of features by N (# of features)


$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

SHapley Additive exPlanations

- Shapley value definition
 - fairly distributes feature importance by considering all possible feature combinations
 - averages feature contributions across all possible coalitions

summing over the subsets

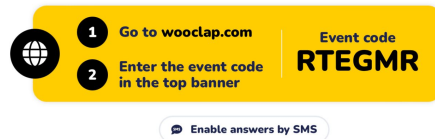
$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

SHapley Additive exPlanations

- SHAP properties
 - **local accuracy**: the sum of the SHAP values equals the model's output minus expected prediction
 - **consistency**: if a feature's impact on the prediction increases, its SHAP value does not decrease
 - **missingness**: features that do not contribute to prediction, have a SHAP value of zero

SHapley Additive exPlanations

- Potential issue?



$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

SHapley Additive exPlanations

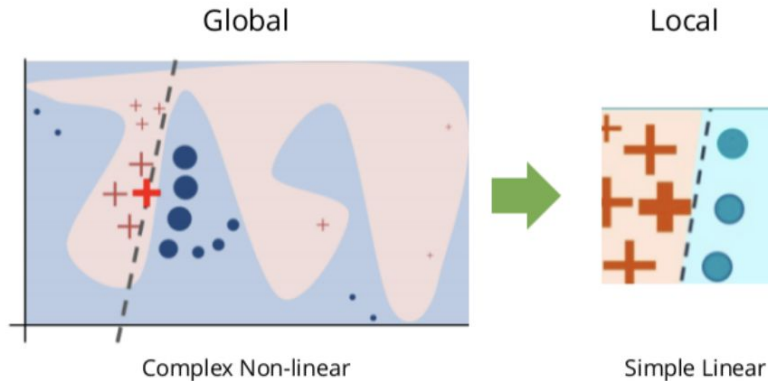
- **Potential issue**

- for N features, SHAP needs to compute 2^N
- as N increases, computations grow exponentially

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)]$$

Back in time - LIME

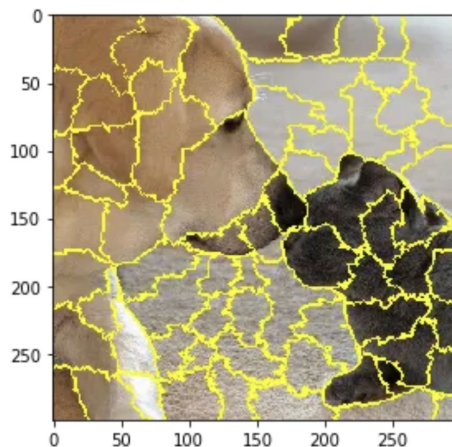
- Local Interpretable Model-agnostic Explanations
 - a **complex model** can be more **easily understood** at a **local level**
 - explain a prediction with the help of an **interpretable surrogate model**



Back in time - LIME

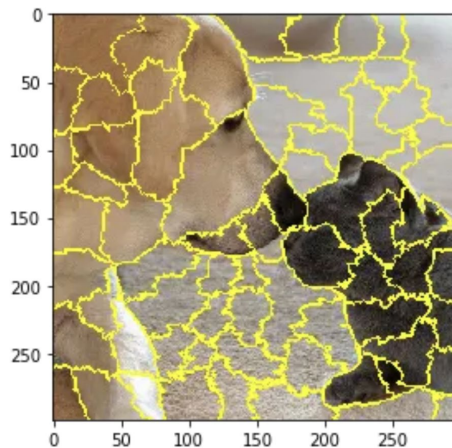
- How LIME works

(1) choose a specific **data point** to **explain**



Back in time - LIME

- How LIME works
 - segmenting **images** into **superpixels**
 - **perturbs** the image by **masking** different **superpixels**

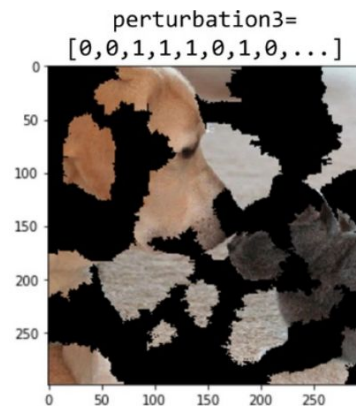
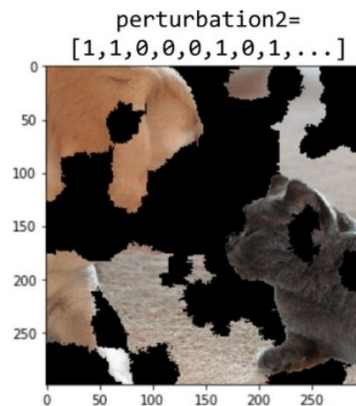
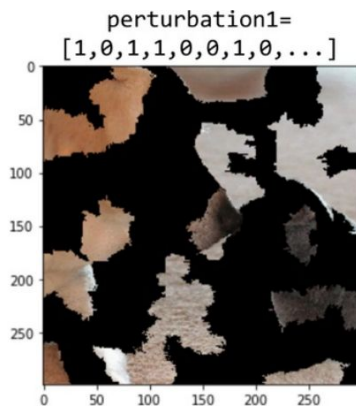
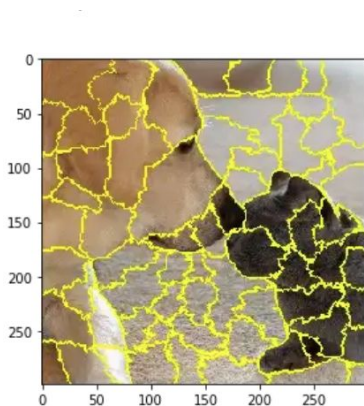


Back in time - LIME

- How LIME works

(1) choose a specific **data point** to **explain**

(2) generate **new synthetic samples** by introducing **perturbations**



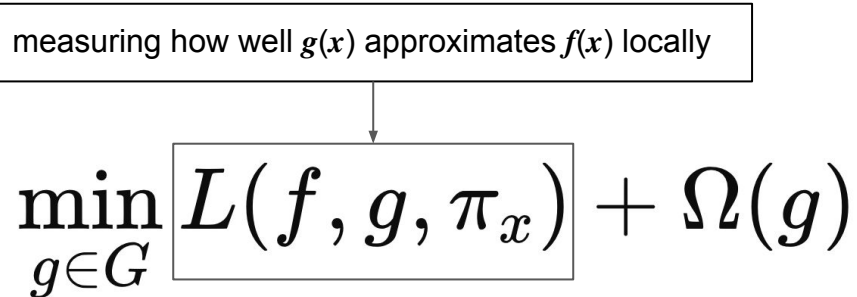
Back in time - LIME

- How LIME works
 - (2) generate **new** synthetic **samples** by introducing **perturbations**
 - (3) **black-box** (complex) model **inference** on **perturbed** samples
 - (4) **train** a (simple) **interpretable model** using the **perturbed samples** and **model outputs**
 - (5) use the **weights** of the **interpretable model** to determine **important features**

Back in time - LIME

- How LIME works
 - approximate the black-box model $f(\mathbf{x})$ with a local interpretable model $g(\mathbf{x})$
 - the interpretable model is only accurate locally

measuring how well $g(x)$ approximates $f(x)$ locally


$$\min_{g \in G} \boxed{L(f, g, \pi_x)} + \Omega(g)$$

Back in time - LIME

- How LIME works
 - approximate the black-box model $f(\mathbf{x})$ with a local interpretable model $g(\mathbf{x})$
 - the interpretable model is only accurate locally

A diagram illustrating the LIME optimization formula. A rectangular box at the top contains the text "complexity penalty to ensure interpretability". A downward-pointing arrow connects this box to the term $\Omega(g)$ in the formula below. The formula is
$$\min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$
 where the entire expression is enclosed in a large rectangular box.

$$\min_{g \in G} L(f, g, \pi_x) + \Omega(g)$$

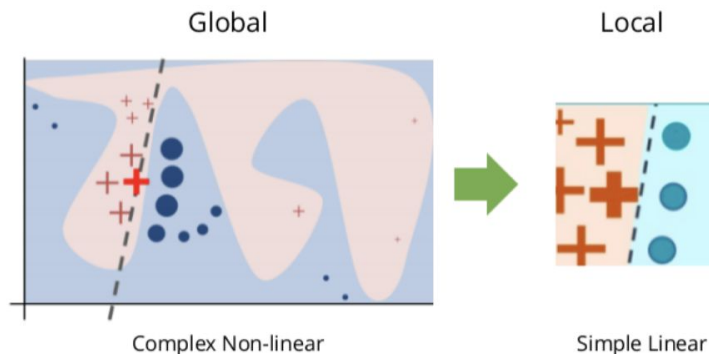
Back in time - LIME

- (Some) limitations of LIME

local accuracy: **does not guarantee** that **sum** of the feature **attributions equals** model's **prediction**

consistency: **does not ensure** that if feature's **contribution increases**, **importance cannot decrease**

symmetry: **may or may not** assign **equal importance** to features with the **same effect** on prediction



SHapley Additive exPlanations

- **KernelSHAP - bridging LIME and SHAP**
 - **kernelSHAP improves** upon **LIME** while **maintaining SHAP's** theoretical **guarantees**
 - **replaces LIME's** arbitrary **weighting function** with **Shapley Kernel**

$$\pi_x(z) = \exp\left(-\frac{D(x, z)^2}{\sigma^2}\right) \longrightarrow \pi(S) = \frac{(M-1)}{\binom{M}{|S|} |S| (M-|S|)}$$

SHapley Additive exPlanations

- KernelSHAP

(1) **randomly sample** feature **subsets** (instead of testing all possible ones)

SHapley Additive exPlanations

- **KernelSHAP**

- (1) randomly sample feature subsets (instead of testing all possible ones)
- (2) **perturb input features** and replace missing values with baseline values

SHapley Additive exPlanations

- **KernelSHAP**

- (1) randomly sample feature subsets (instead of testing all possible ones)
- (2) perturb input features and replace missing values with baseline values
- (3) run the **model** on **perturbed instances** to measure changes in predictions

SHapley Additive exPlanations

- **KernelSHAP**

- (1) randomly sample feature subsets (instead of testing all possible ones)
- (2) perturb input features and replace missing values with baseline values
- (3) run the model on perturbed instances to measure changes in predictions
- (4) **weight** each perturbed **sample** using the **Shapley kernel**

$$\pi(S) = \frac{(M - 1)}{\binom{M}{|S|} |S| (M - |S|)}$$

SHapley Additive exPlanations

- **KernelSHAP**

- (1) randomly sample feature subsets (instead of testing all possible ones)
- (2) perturb input features and replace missing values with baseline values
- (3) run the model on perturbed instances to measure changes in predictions
- (4) weight each perturbed sample using the Shapley kernel
- (5) **fit a weighted linear regression model** to estimate SHAP values

$$\hat{g}(x) = \beta_0 + \sum_{i=1}^M \beta_i x_i$$

SHapley Additive exPlanations

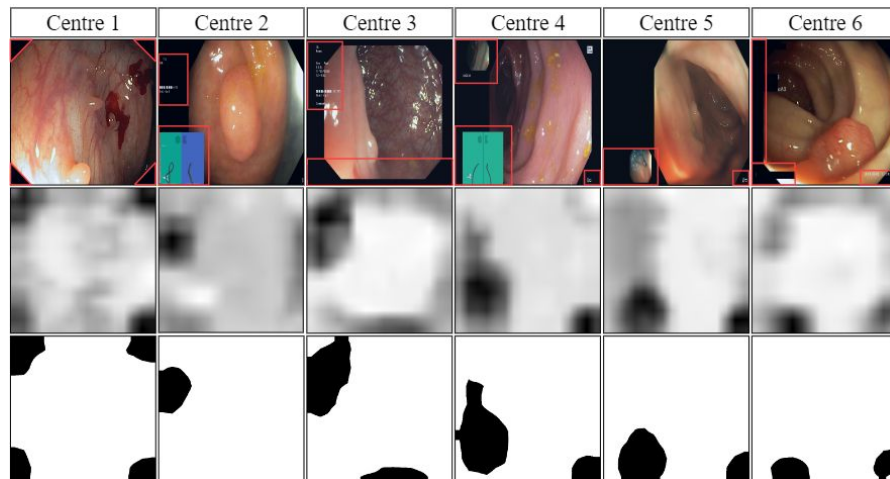
- **KernelSHAP**

- (1) randomly sample feature subsets (instead of testing all possible ones)
- (2) perturb input features and replace missing values with baseline values
- (3) run the model on perturbed instances to measure changes in predictions
- (4) weight each perturbed sample using the Shapley kernel
- (5) fit a weighted linear regression model to estimate SHAP values
- (6) extract **feature importance** and generate SHAP-based explanations

Saliency maps beyond model understanding



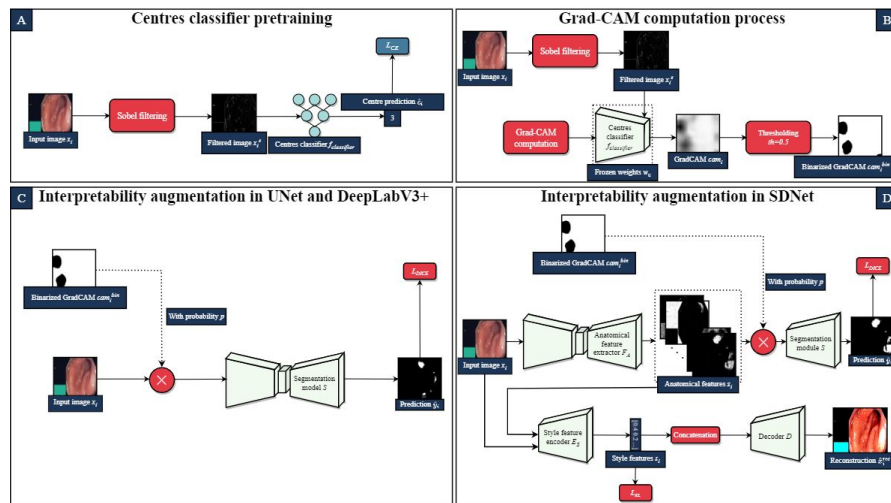
- Robust and Explainable Segmentation Models
 - **Domain shifts** lead to drops in performance
 - **Disentangled** representation learning
 - New **data augmentation** strategies



Saliency maps beyond model understanding



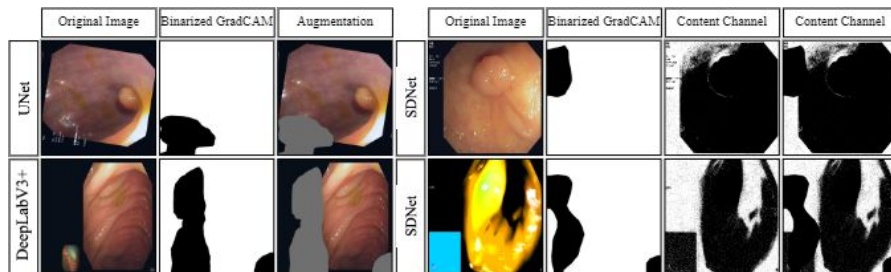
- Robust and Explainable Segmentation Models
 - Innovative **data augmentation**
 - Center-specific **saliency maps**
 - **Focus** on **anatomical** regions



Saliency maps beyond model understanding



- Robust and Explainable Segmentation Models
 - **Multiply** the original **image** by **saliency map**
 - Probabilistic approach (probability p)
 - **DRL**: multiplication in **anatomical** representation

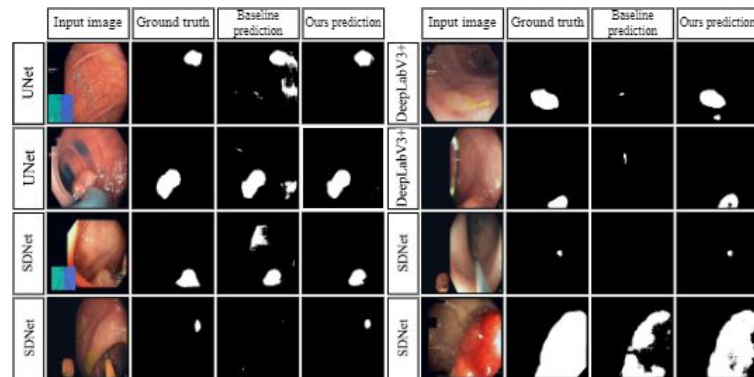


Saliency maps beyond model understanding

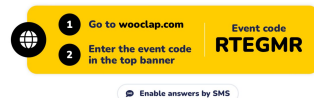


- Robust and Explainable Segmentation Models
 - **Reduced noise levels**, better at **small polyps**, overall **improved performance**

	Dice	Recall	Accuracy	Dice	Recall	Accuracy
Out-dist set	UNet			UNet interpretability augmentation		
centre 1	0.7257	0.7725	0.9589	0.7353 (60%)	0.7716	0.9587
centre 2	0.5762	0.6454	0.9440	0.6062 (60%)	0.5948	0.9530
centre 3	0.7054	0.6724	0.9619	0.7470 (40%)	0.6966	0.9658
centre 4	0.4223	0.3804	0.9594	0.4519 (40%)	0.3757	0.9607
centre 5	0.4725	0.4934	0.9581	0.4893 (40%)	0.5369	0.9583
centre 6	0.6423	0.5597	0.9593	0.6574 (60%)	0.6132	0.9634
	Dice	Recall	Accuracy	Dice	Recall	Accuracy
Out-dist set	Deeplabv3+			Deeplabv3+ interpretability augmentation		
centre 1	0.6003	0.6284	0.9482	0.6155 (60%)	0.6195	0.9545
centre 2	0.5498	0.5358	0.9398	0.5679 (60%)	0.5159	0.9496
centre 3	0.5696	0.5407	0.9585	0.6314 (40%)	0.6745	0.9561
centre 4	0.3424	0.2471	0.9483	0.3592 (50%)	0.2166	0.9493
centre 5	0.3867	0.4171	0.9543	0.4046 (60%)	0.4479	0.9533
centre 6	0.6268	0.5932	0.9631	0.6342 (40%)	0.6126	0.9658
	Dice	Recall	Accuracy	Dice	Recall	Accuracy
Out-dist set	SDNet			SDNet interpretability augmentation		
centre 1	0.7130	0.7583	0.9551	0.7226 (40%)	0.7726	0.9575
centre 2	0.5489	0.5794	0.9328	0.5579 (50%)	0.5482	0.9464
centre 3	0.7151	0.7082	0.9620	0.7208 (40%)	0.7245	0.9603
centre 4	0.3981	0.3254	0.9591	0.3841 (50%)	0.3360	0.9557
centre 5	0.4312	0.4293	0.9587	0.4546 (40%)	0.4722	0.9588
centre 6	0.6398	0.6195	0.9610	0.6626 (60%)	0.5977	0.9621



Saliency maps beyond model understanding

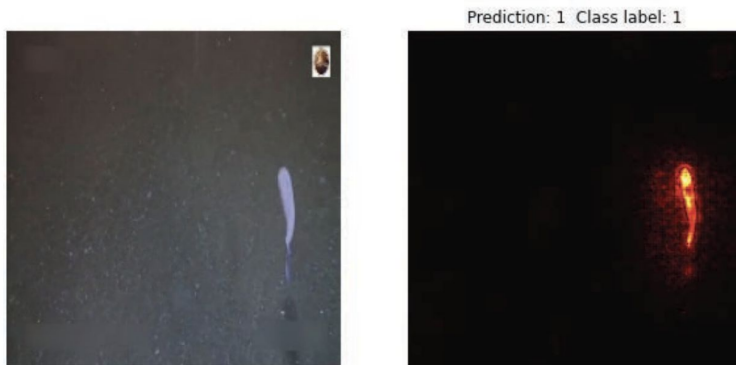


- Weakly supervised object detection
 - how to use **saliency maps** to perform **weakly-supervised object detection**



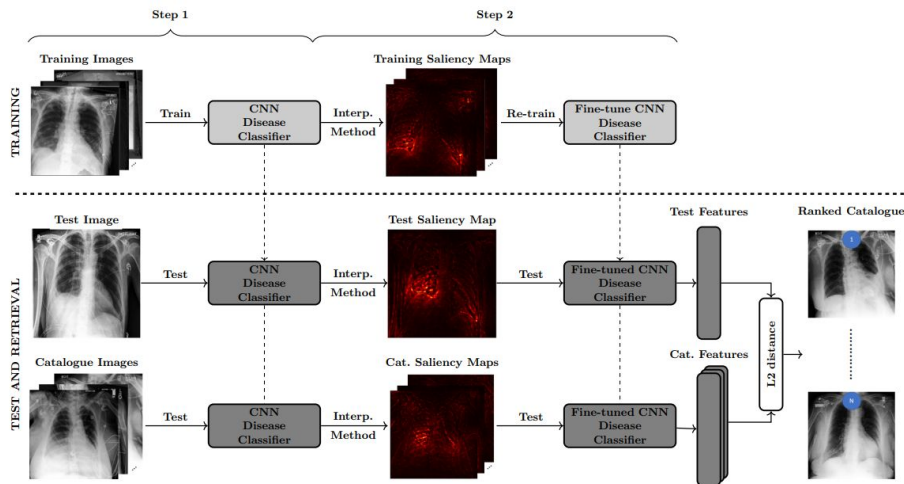
Saliency maps beyond model understanding

- Weakly supervised object detection
 - using **saliency maps** as a proxy to **annotate** fish images



Saliency maps beyond model understanding

- **Enhance the retrieval** of medical images (more details in next lecture)



Layer-wise Relevance Propagation

AI Model – Explainability Method		
Generation (method)	Ante-hoc/Post-hoc	
	Model-specific/Model-agnostic	
	Scope (Local/global)	





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SHapley Additive exPlanations

AI Model – Explainability Method		
Generation (method)	Ante-hoc/Post-hoc	
	Model-specific/Model-agnostic	
	Scope (Local/global)	





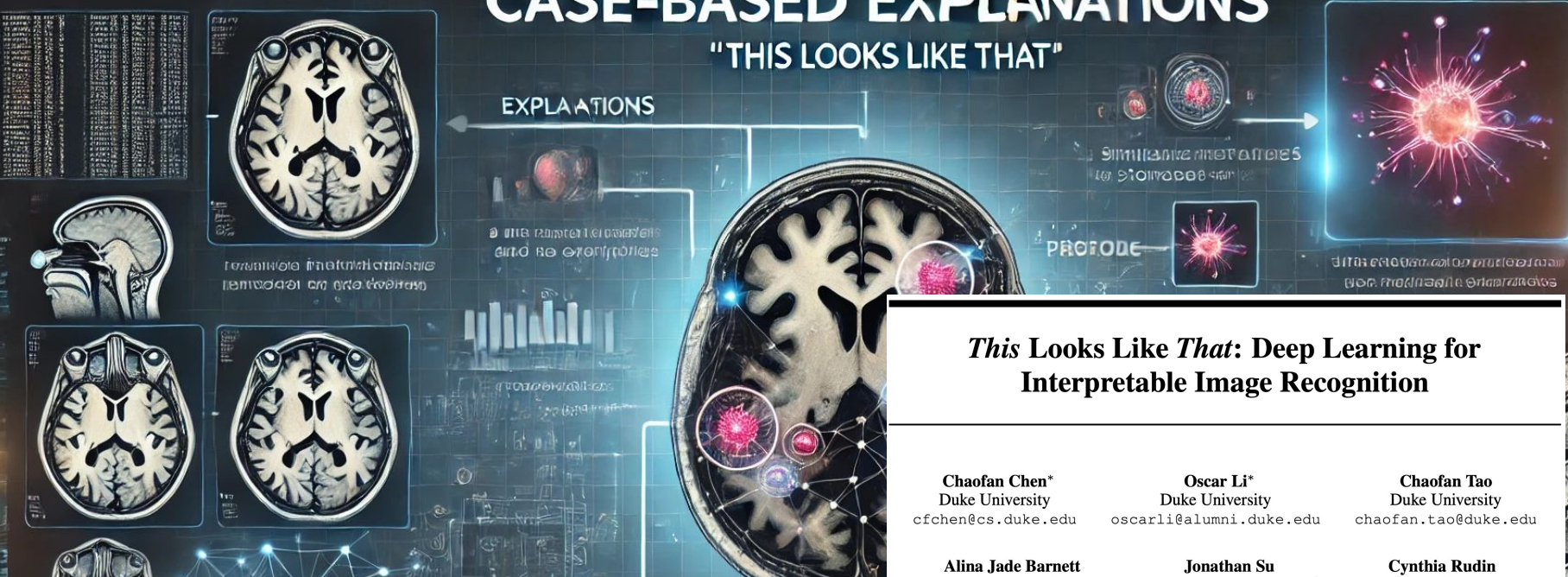
- 1 Go to wooclap.com
- 2 Enter the event code in the top banner

Event code
NMZJKY

 Enable answers by SMS

CASE-BASED EXPLANATIONS

"THIS LOOKS LIKE THAT"



This Looks Like That: Deep Learning for Interpretable Image Recognition

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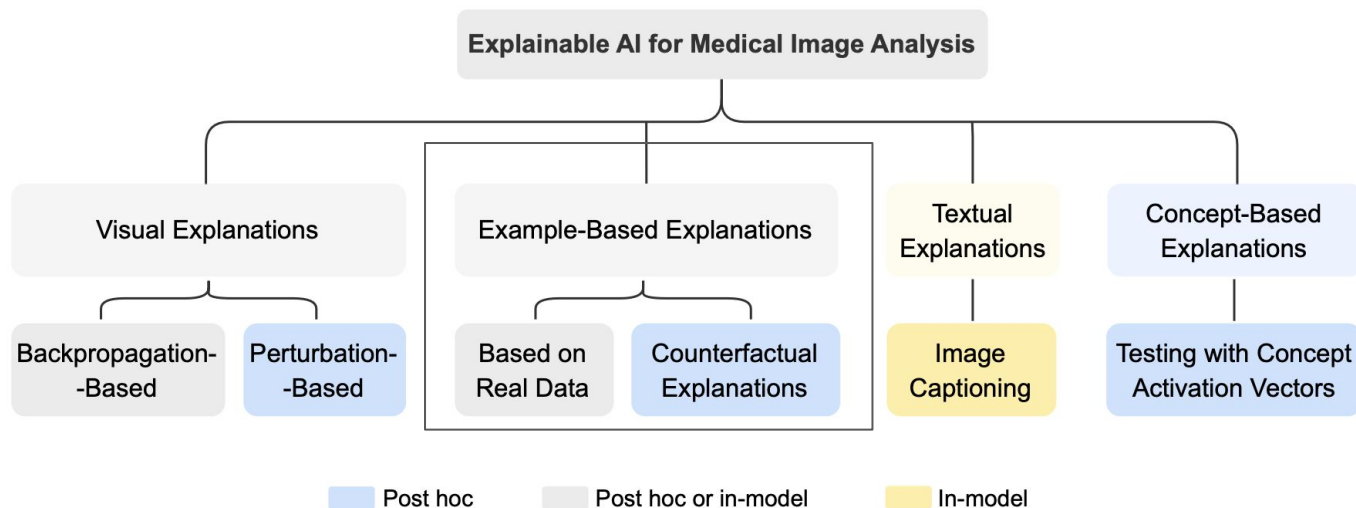
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Example-based Explanations

Explainable ML (for Medical Image Analysis)



Explainable ML (for Medical Image Analysis)

- **Example-based** explanations
 - (or **case-based**) explanations **present past cases** to **justify** model's **prediction**
 - **aligns** with medical practice - **radiologists** often look for **similar cases** in databases

Case of the Day



Annular pancreas

Contributed by [Sepehr Haghighi](#)

This case demonstrates the typical appearance of an [annular pancreas](#), a morphological anomaly in which pancreatic tissue encircles the [duodenum](#). The condition is divided into:

- complete annular pancreas: pancreatic parenchyma or an annular duct is seen to surround the 2nd part of the duodenum completely
- incomplete annular pancreas...

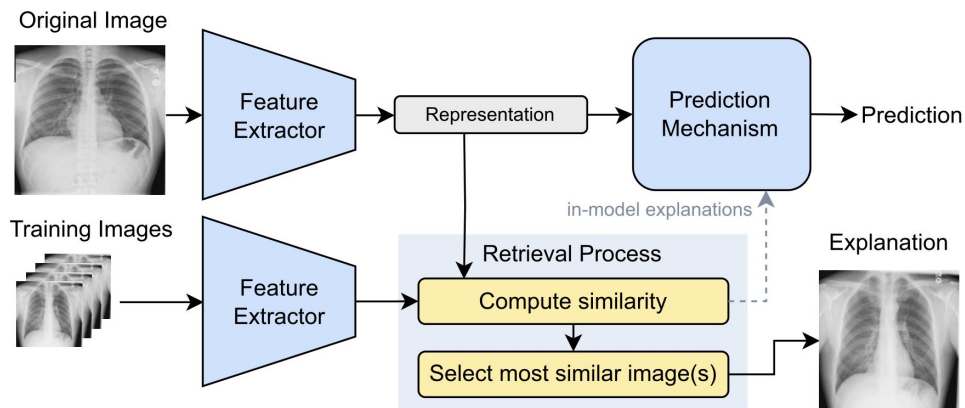
[View Previous Cases](#) [VIEW CASE](#)

Example-based Explanations

- Types of **example-based** explanations:
 - based on **real data** - derived from actual patient cases
 - **counterfactual** explanations - uses synthetic data to show *what-if* scenarios

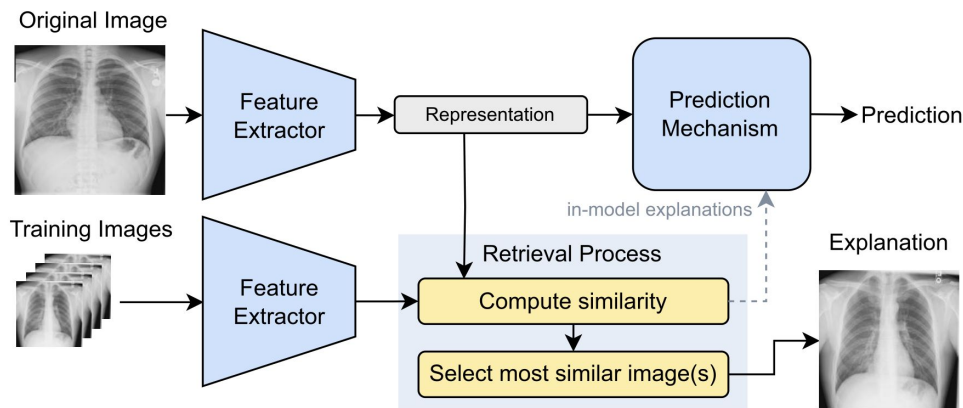
Explanations based on Real Data

- Explanations based on Real Data:
 - via **intrinsically interpretable** models
 - or in a **post hoc** manner



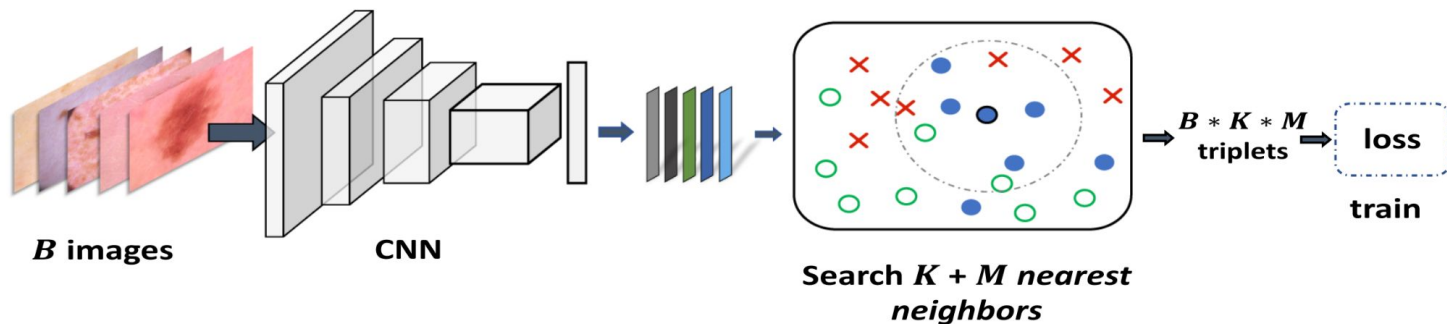
Explanations based on Real Data

- **In-model approaches** (directly use **training images** to make a **decision** about **test image**):
 - **prototype-based** (classify images based on **similarity to learned prototypes**)
 - **distance-based** (map images to a **semantic latent space** and classify based on **distance**)



Explanations based on Real Data

- **DeepKNN** (distance-based)
 - **promotes** that **each** training **sample** and its **nearest neighbors** are of the **same class**
 - learns a **meaningful latent space** where **similar images cluster** together



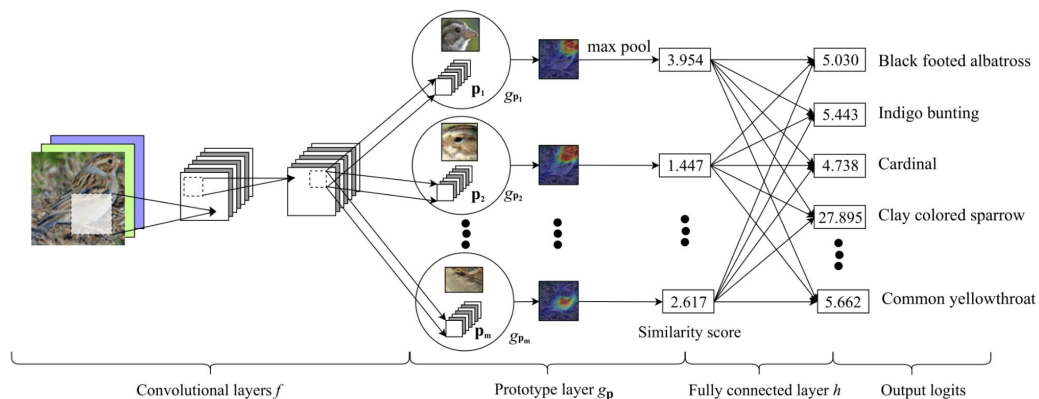
Explanations based on Real Data

- **DeepKNN** (distance-based)
 - helps **improve performance** on small-class and **imbalanced** image datasets
 - **predictions** are **explained** by retrieving the **closest training samples**

Datasets	Skin-7			Pneumonia			SD-198		
Methods	Acc	MCR	RS	Acc	MCR	RS	Acc	MCR	RS
kNN (VGG19, $K_p = 1$) [17]	71.37	42.03	37.93	52.67	52.36	43.06	24.49	24.37	20.83
kNN (ResNet50, $K_p = 1$) [18]	75.56	49.92	55.17	52.37	51.24	38.90	27.62	28.38	29.16
Triplet (ResNet50, $K_p = 1$) [12]	82.91	64.21	63.21	67.84	66.32	65.21	59.96	60.05	47.31
Deep kNN (ours, $K_p = 1$)	88.2	77.4	77.0	71.0	69.3	67.3	64.5	64.1	48.3
kNN (VGG19, $K_p = 5$) [17]	70.72	30.77	10.34	55.42	55.71	44.97	15.12	13.28	4.17
kNN (ResNet50, $K_p = 5$) [18]	74.41	37.53	34.48	55.52	54.66	39.65	19.11	17.45	12.50
Triplet (ResNet50, $K_p = 5$) [12]	84.29	68.31	67.34	70.00	68.73	66.32	60.17	60.02	47.21
Deep kNN (ours, $K_p = 5$)	89.1	78.9	77.3	71.1	69.4	69.0	65.1	64.3	48.3
Weighted-CE (ResNet50) [19]	88.02	80.21	76.60	71.12	69.11	70.05	61.90	62.40	47.67
Deep kNN* (ours, $K_p = 5$)	90.3	81.0	80.4	71.6	71.1	70.9	66.4	66.4	51.5

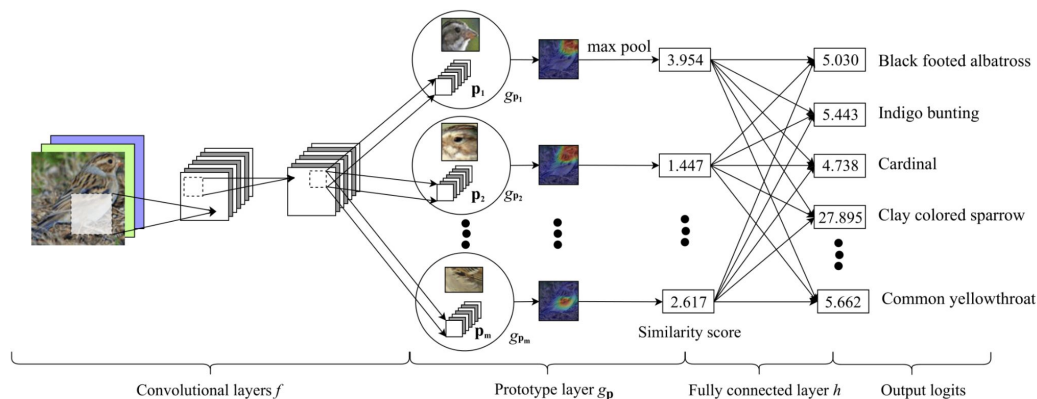
Explanations based on Real Data

- **ProtoPNet** (prototype-based)
 - network dissects the image by **finding prototypical** parts
 - **combines** evidence from the **prototypes** to make a final **classification**



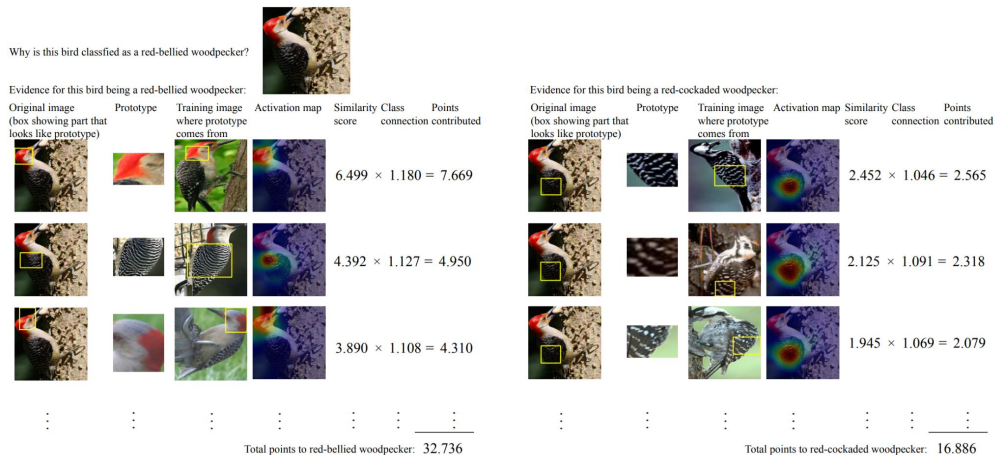
Explanations based on Real Data

- **ProtoPNet** (prototype-based)
 - penalizes **misclassification** on the training data (cross entropy loss)
 - minimization of **cluster cost** (latent patch close to prototype of its own class)
 - minimization of **separation cost** (latent patch to stay away from prototypes of other classes)



Explanations based on Real Data

- **ProtoPNet** (prototype-based)
 - the model computes its **latent representation** $f(x)$
 - searches for evidence by **comparing** $f(x)$ to all **prototypes** of a specific **class**



Explanations based on Real Data

- **ProtoPNet**
 - maintains **comparable accuracy** to its corresponding **non-interpretable baseline**
 - provides **interpretability** with **minimal trade-off** in classification **performance**

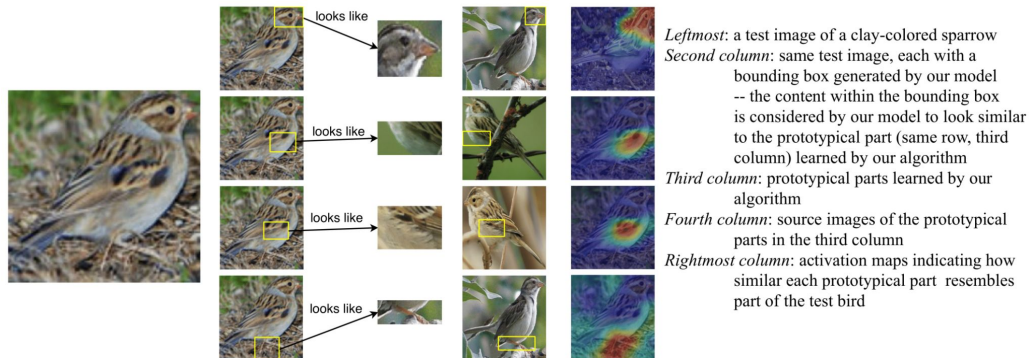
Table 1: Top: Accuracy comparison on cropped bird images of CUB-200-2011
Bottom: Comparison of our model with other deep models

Base	ProtoPNet	Baseline	Base	ProtoPNet	Baseline
VGG16	76.1 ± 0.2	74.6 ± 0.2	VGG19	78.0 ± 0.2	75.1 ± 0.4
Res34	79.2 ± 0.1	82.3 ± 0.3	Res152	78.0 ± 0.3	81.5 ± 0.4
Dense121	80.2 ± 0.2	80.5 ± 0.1	Dense161	80.1 ± 0.3	82.2 ± 0.2

Explanations based on Real Data

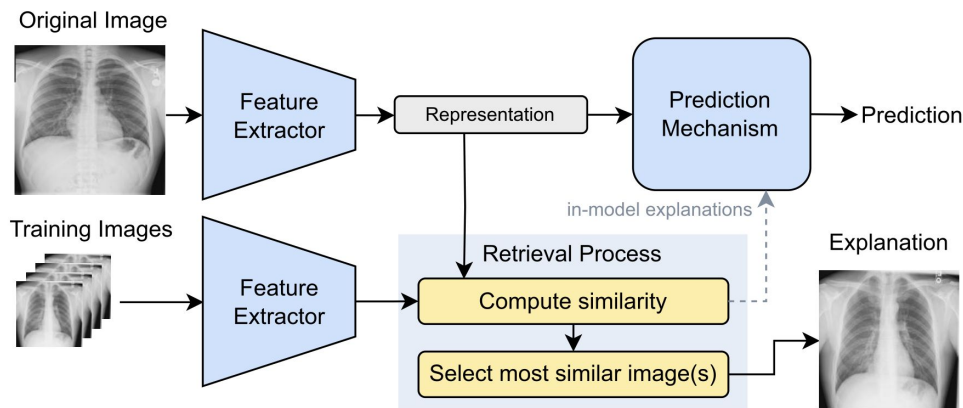
- **ProtoPNet**

- combines deep learning with case-based reasoning for interpretability
- provides prototype-based explanations
- unlike black-box CNNs, makes decisions based on real understandable features



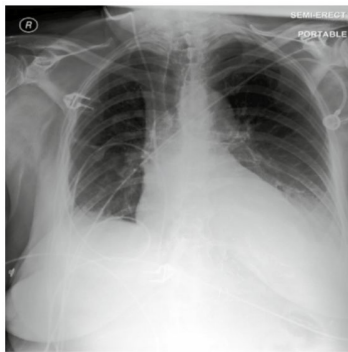
Explanations based on Real Data

- **Post hoc** example-based explanations
 - **retrieve** the **closest training samples** as explanations (**similarity** in the **latent space**)
 - **classify images** based on **latent representation** rather than distance to training samples

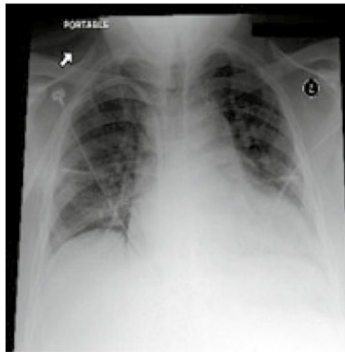


Explanations based on Real Data

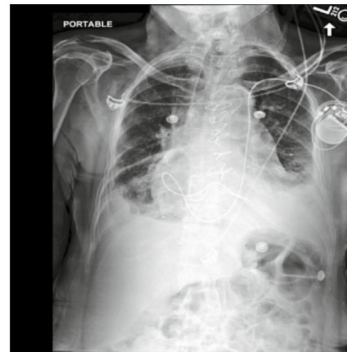
- **Post hoc** example-based explanations
 - asked a board-certified **radiologist** to **rank images** in terms of **disease similarity**
 - **disease similarity** is **different** from **image similarity**



(a) Test Image



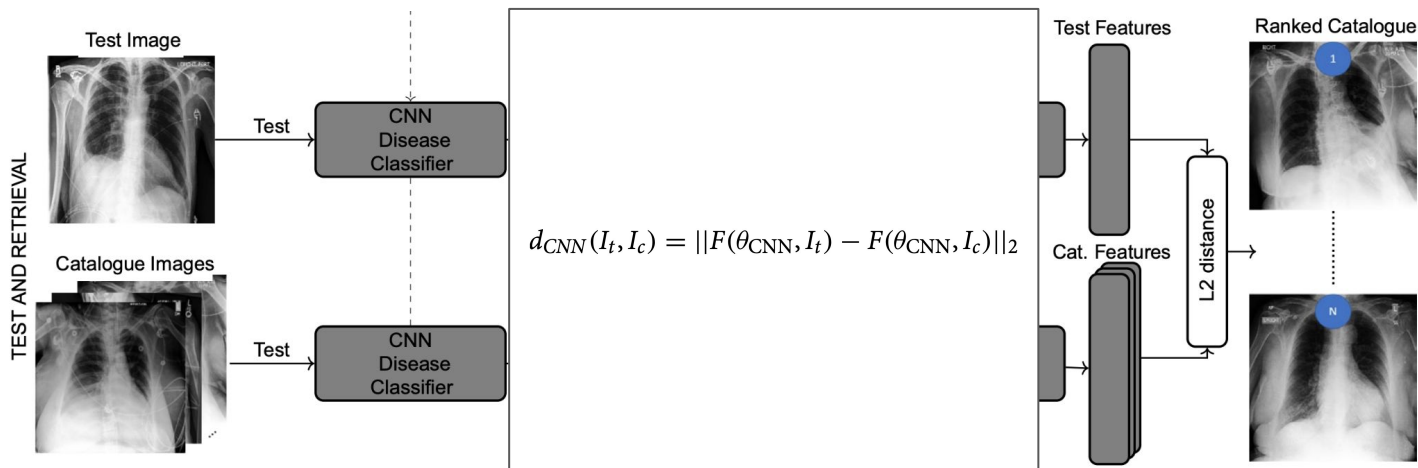
(b) Least Similar Image



(c) Most Similar Image

Explanations based on Real Data

- **Post hoc** example-based explanations
 - **CNN-based disease** classifier extracts **semantic** representations
 - **similarity** is **computed** in the **semantic space** close to the decision

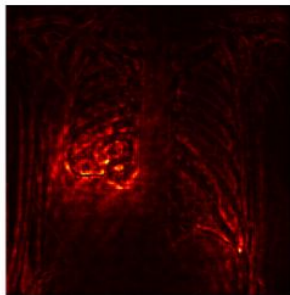


Explanations based on Real Data

- **Post hoc** example-based explanations
 - need to **focus representations** in **disease** relevant **regions** for improved retrieval
 - **visual explanations** allow finding **relevant regions without** (strong) **supervision**



Chest X-ray Image



Saliency Map

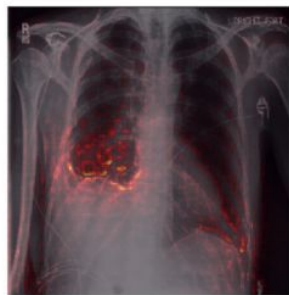
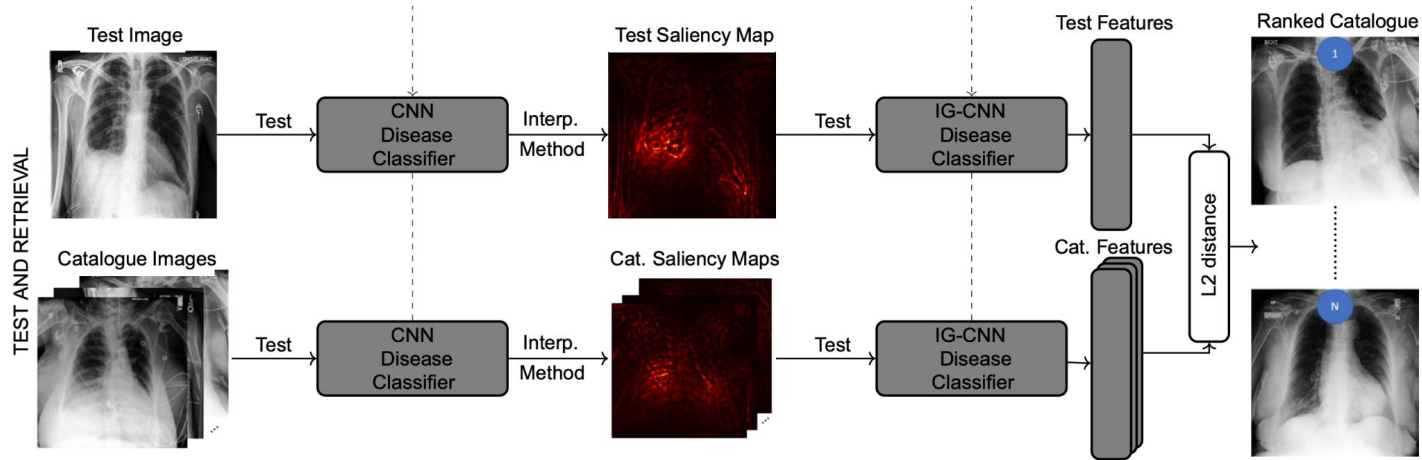


Image with Saliency

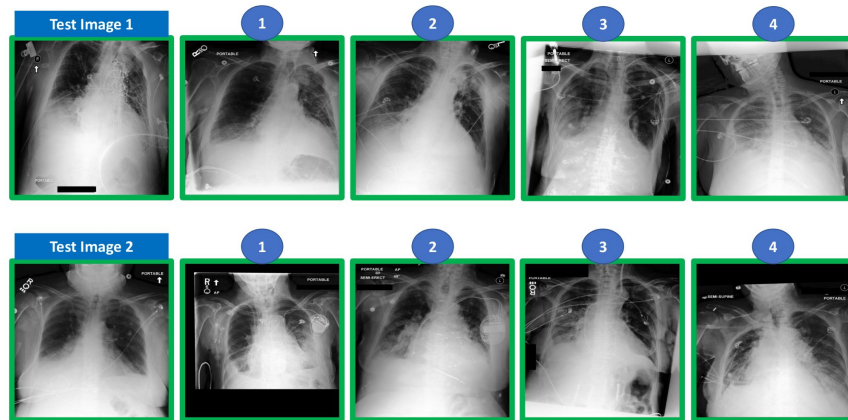
Explanations based on Real Data

- **Post hoc** example-based explanations
 - two-step training process
 - **saliency maps** are used to **fine-tune CNN disease classifier**



Explanations based on Real Data

- **Post hoc** example-based explanations
 - test data is split into query and catalogue of 10 images
 - catalogue is ranked by the radiologist



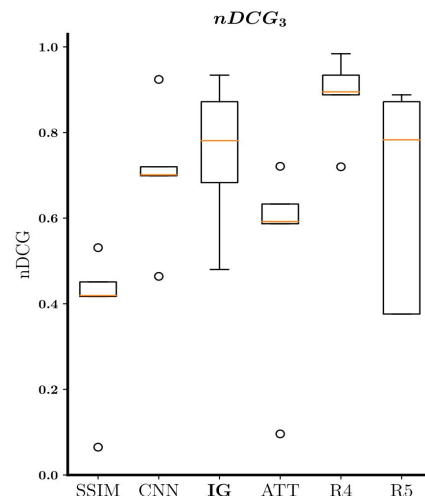
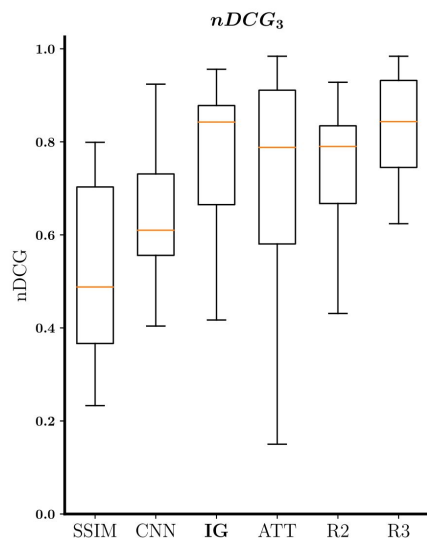
Explanations based on Real Data

- **Post hoc** example-based explanations
 - **SSIM** - structural similarity index (statistically-based image similarity)
 - **CNN** - deep learning baseline (task dependent)
 - **CNN + attention layers** (another approach to focus representations)
 - Performance compared with two additional radiologists (inter-rater variability)

$$\text{nDCG}_p = \frac{DCG_p}{IDCG_p}$$

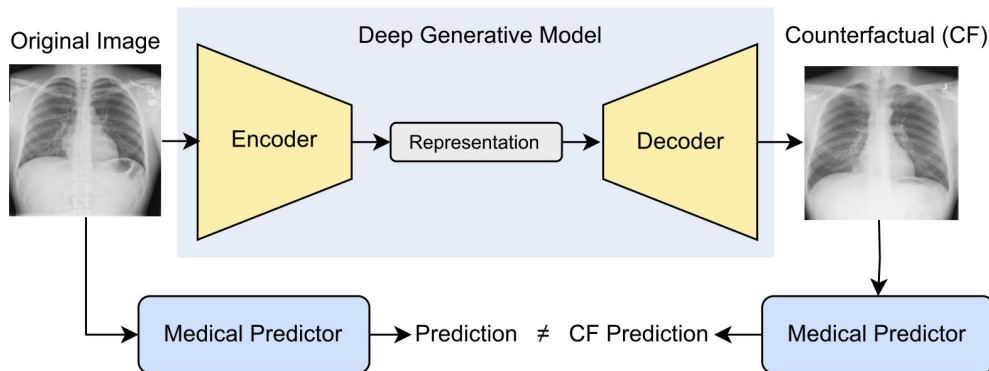
$$DCG_p = \sum_{i=1}^p \frac{2^{rel_i} - 1}{\log_2(i + 1)}$$

Explanations based on Real Data



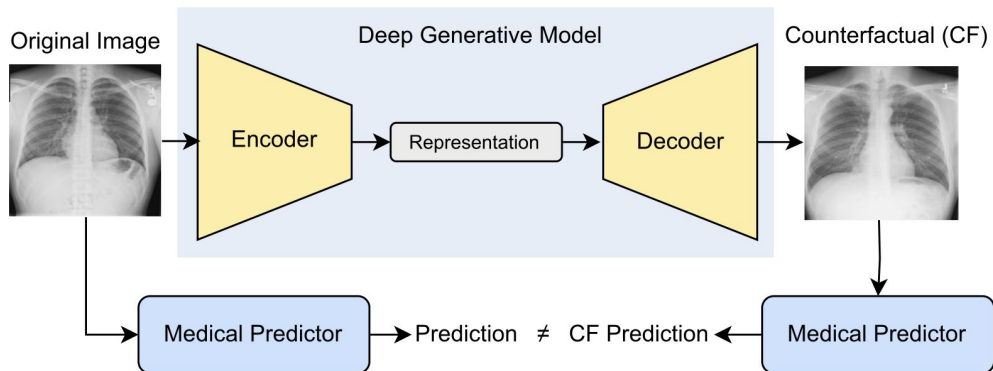
Counterfactual Explanations

- Learning from what-if scenarios
 - **synthetic images** showing the **minimal changes** needed to **alter a model's prediction**
 - **deep generative models** create realistic variations of the test image with a different classification
 - helps understand decision boundaries and key features influencing predictions



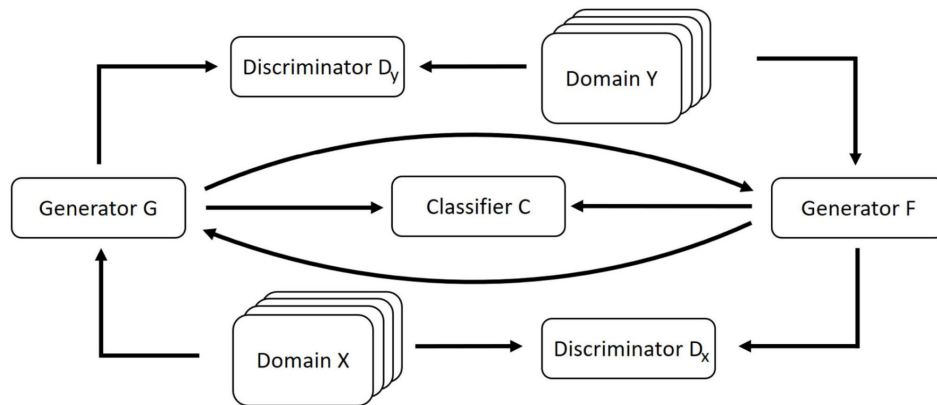
Counterfactual Explanations

- Learning from what-if scenarios
 - **counterfactual explanations** must **modify regions** causally related to the medical **task**
 - preventing the generation of **adversarial samples** is challenging



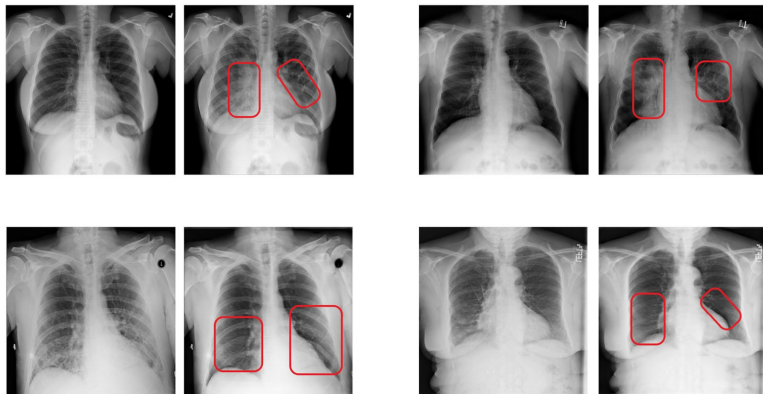
Counterfactual Explanations

- Learning from what-if scenarios
 - takes an input medical image (e.g., chest X-ray)
 - uses CycleGAN to generate the counterfactual
 - ensures new image is realistic and helps understand what features drive the decision



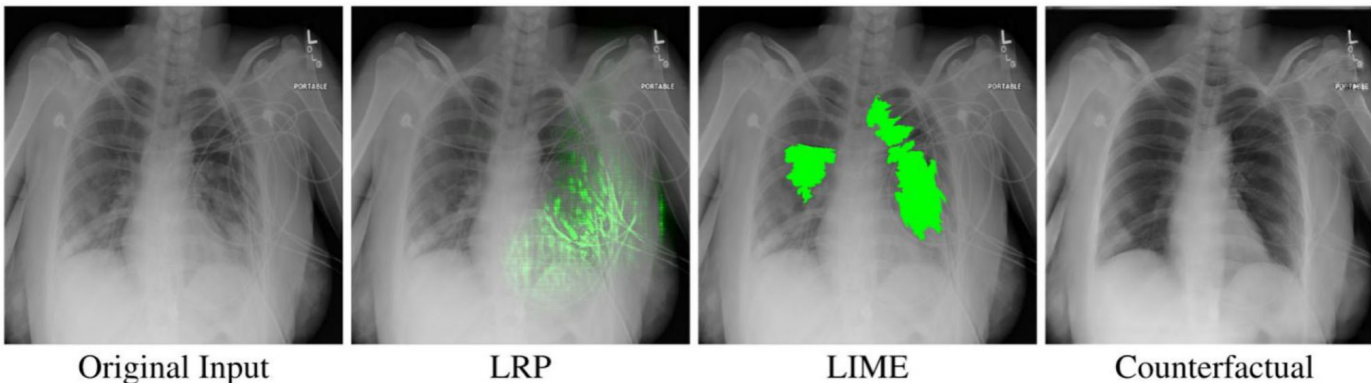
Counterfactual Explanations

- Learning from what-if scenarios
 - CycleGAN allows image-to-image translation without requiring paired images
 - counterfactual loss for decision change (generated image flips the prediction)
 - identity loss (input images stay the same, if they already belong to target domain)



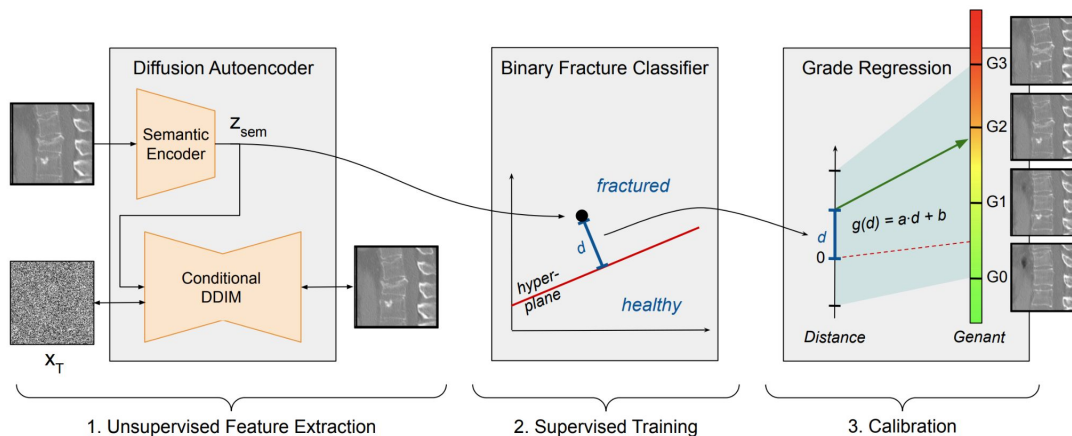
Counterfactual Explanations

- Learning from what-if scenarios
 - counterfactuals show how input image would need to change to alter decision
 - consumers do not have to guess the meaning of the highlighted areas, unlike saliency maps
 - counterfactuals provide a clear, example-based explanation rather than feature importance

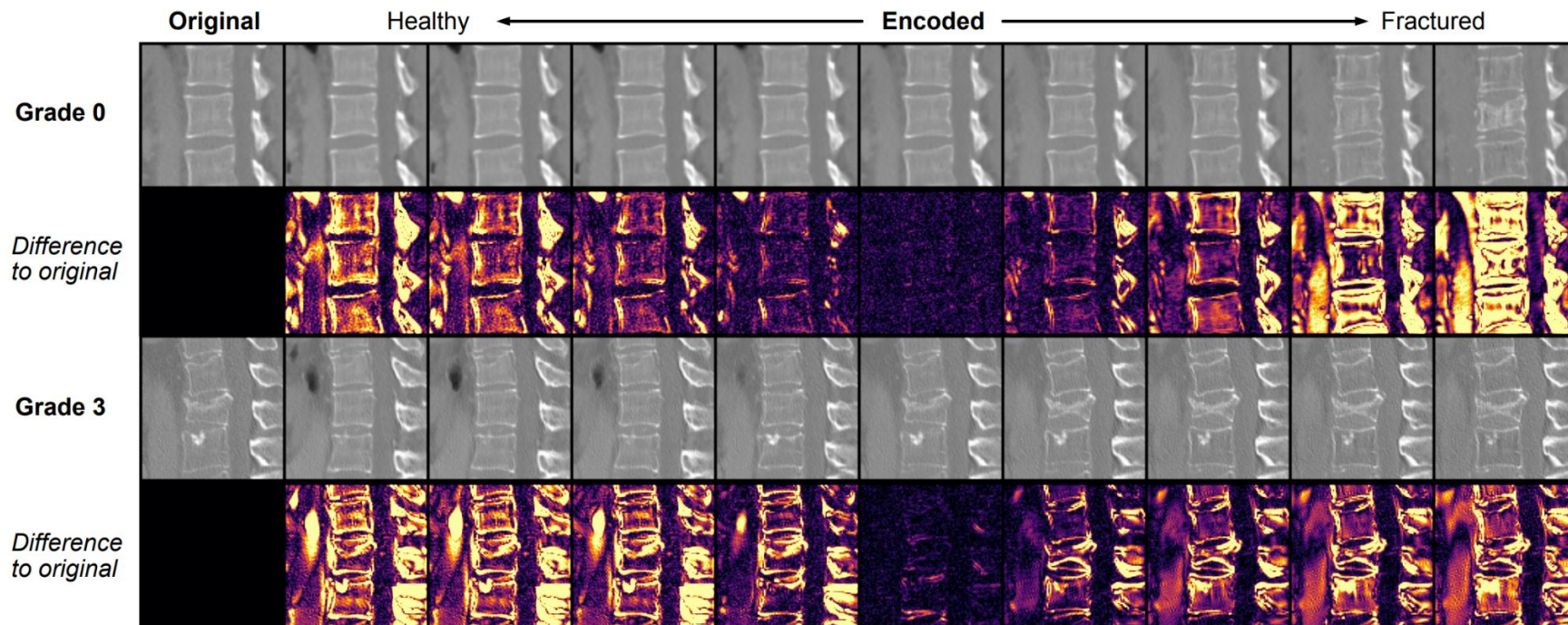


Counterfactual Explanations

- Learning from what-if scenarios
 - unsupervised training of feature extractor using a Diffusion Autoencoder
 - supervised training using a binary classifier (G0 vs. G2/3) to detect fractures
 - calibrating a linear regression of the Genant grade



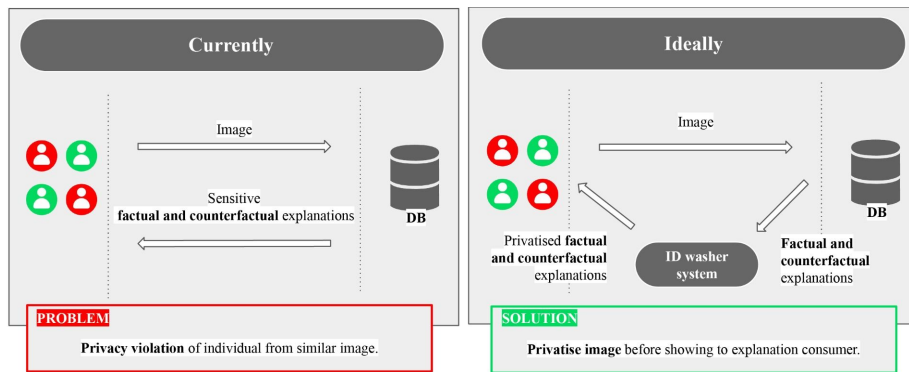
Counterfactual Explanations



Beyond conventional example-based



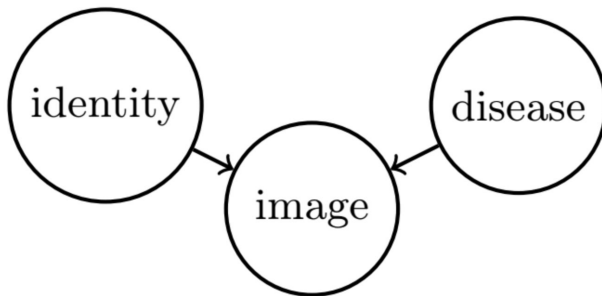
- Privacy-preserving Case-based Explanations
 - Case-based explanations may **expose identity**
 - **Visual anonymization is needed before sharing** (e.g., medical apps, fed. learning)



Beyond conventional example-based



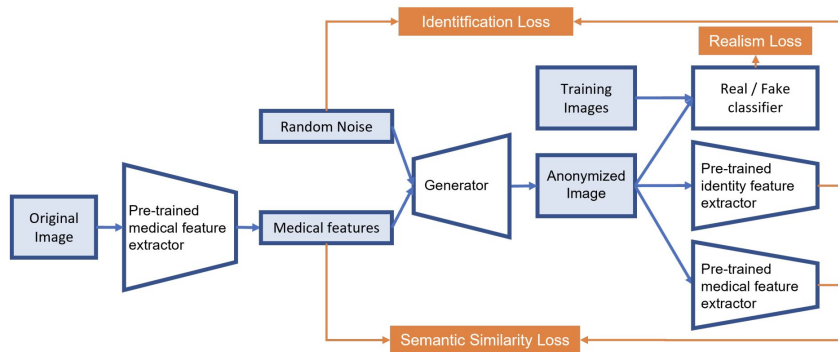
- Privacy-preserving Case-based Explanations
 - Generative Adversarial Network
 - Disentangle identity and medical features
 - Inherently capable of generating counterfactuals



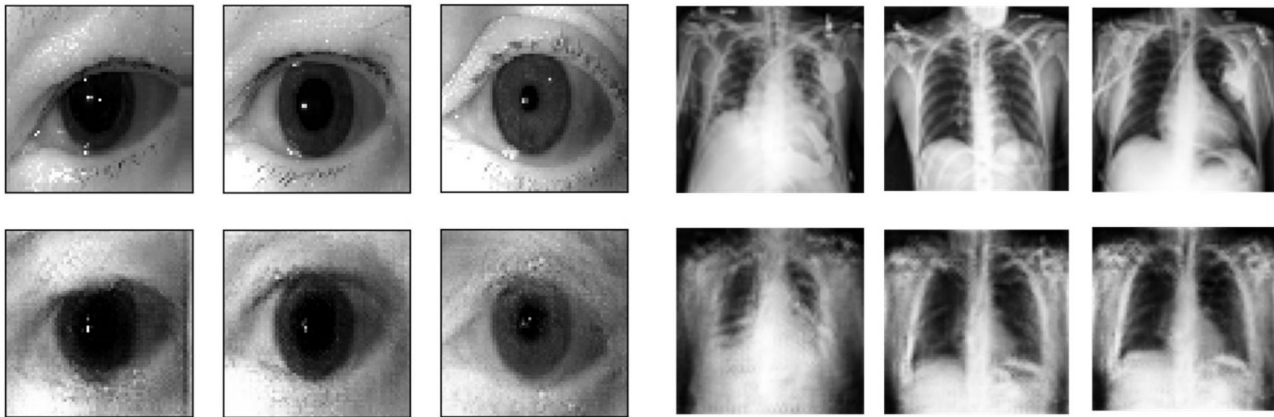
Beyond conventional example-based



- Privacy-preserving Case-based Explanations
 - generative module (generation of the anonymized image)
 - identity module (promotes the association between random noise and the output identity)
 - explanatory module (promotes the preservation of explanatory evidence)



Beyond conventional example-based

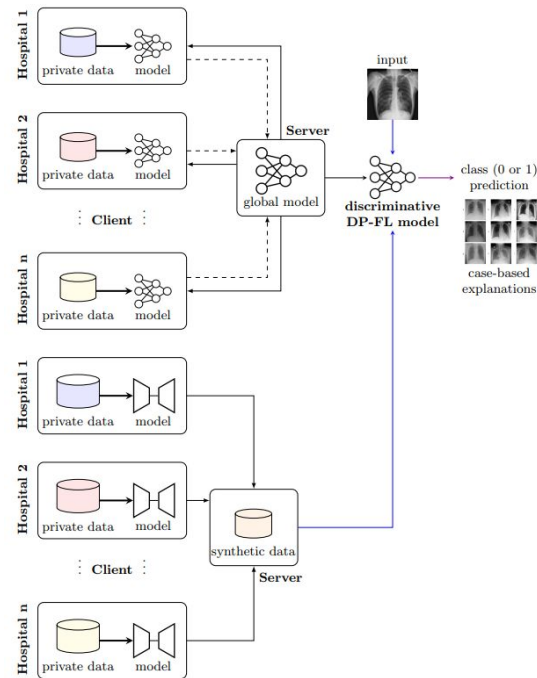


Model	Identity Recognition		Disease Recognition	
	Accuracy	Max Avg Score	Accuracy	F-score
Iris data ($\lambda_{exp} = 3, \lambda_{id} = 5$)	0.88%	13.22%	87.06%	75.82%
Iris data (PP-SIR from [18])	1.18%	2.09%	74.71%	57.84%
Chest data ($\lambda_{exp} = 5, \lambda_{id} = 5$)	37.33%	51.16%	79.40%	82.08%

Beyond conventional example-based



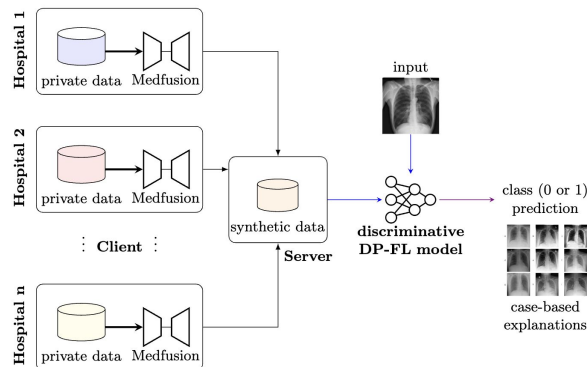
- Privacy-preserving Case-based Explanations
 - **explaining decisions** in the context of **Federated Learning**
 - discriminative federated model
 - generative federated model



Beyond conventional example-based



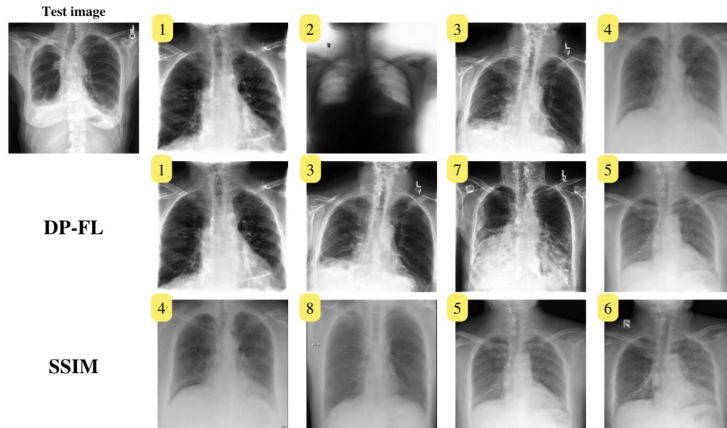
- Privacy-preserving Case-based Explanations
 - three most similar **synthetic images** for each **client**
 - **distance** between **test image** and pool of **synthetic images**



Beyond conventional example-based



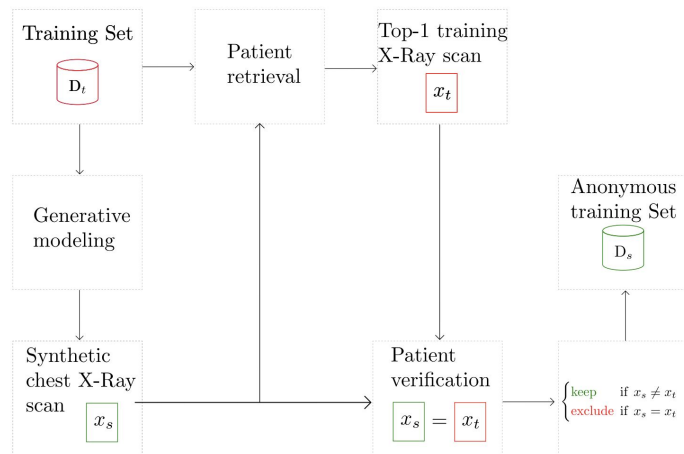
- Privacy-preserving Case-based Explanations
 - top-4 retrieved images obtained by model in comparison with radiologist and SSIM



Beyond conventional example-based



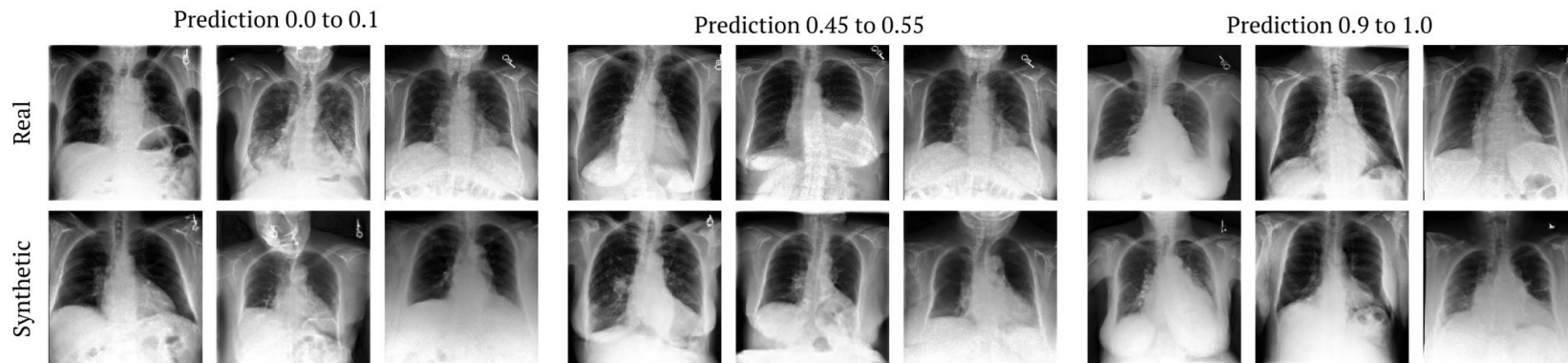
- Privacy-preserving Case-based Explanations
 - use **latent diffusion models** to generate high-quality **synthetic images**
 - **filter out** images that **resemble** real **patients** to ensure privacy
 - retrieve explanations from an anonymous synthetic dataset



Beyond conventional example-based



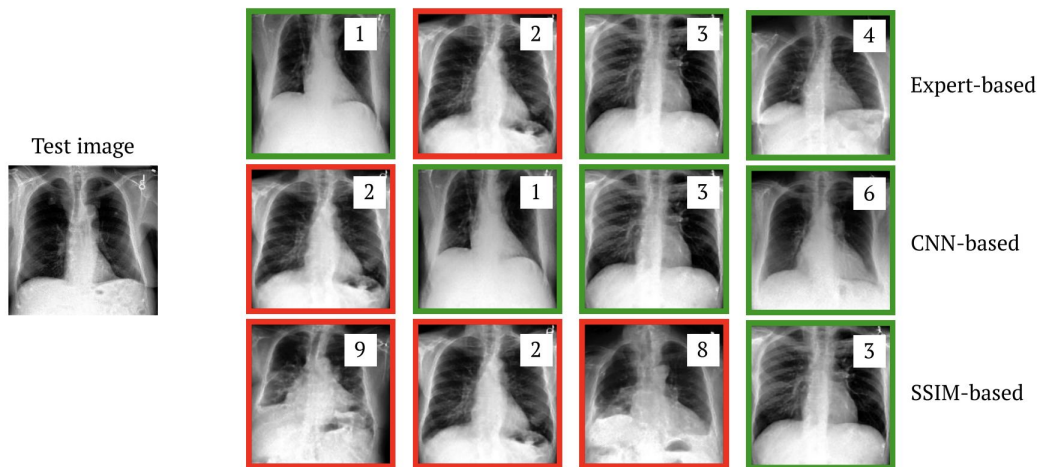
- Privacy-preserving Case-based Explanations
 - verification network classifies whether two images belong to the same patient
 - trained on positive pairs (same patient) and negative pairs (different patients)



Beyond conventional example-based

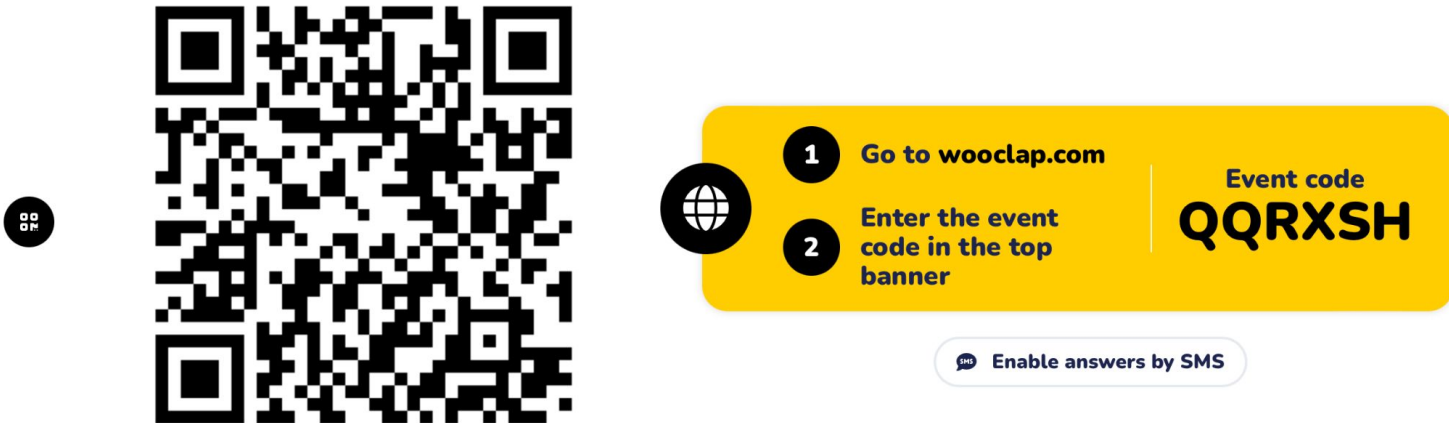


- Privacy-preserving Case-based Explanations
 - generative model can **produce images** with a **specific diagnosis**
 - synthetic images are just **as relevant as** real ones, supporting their use as explanations

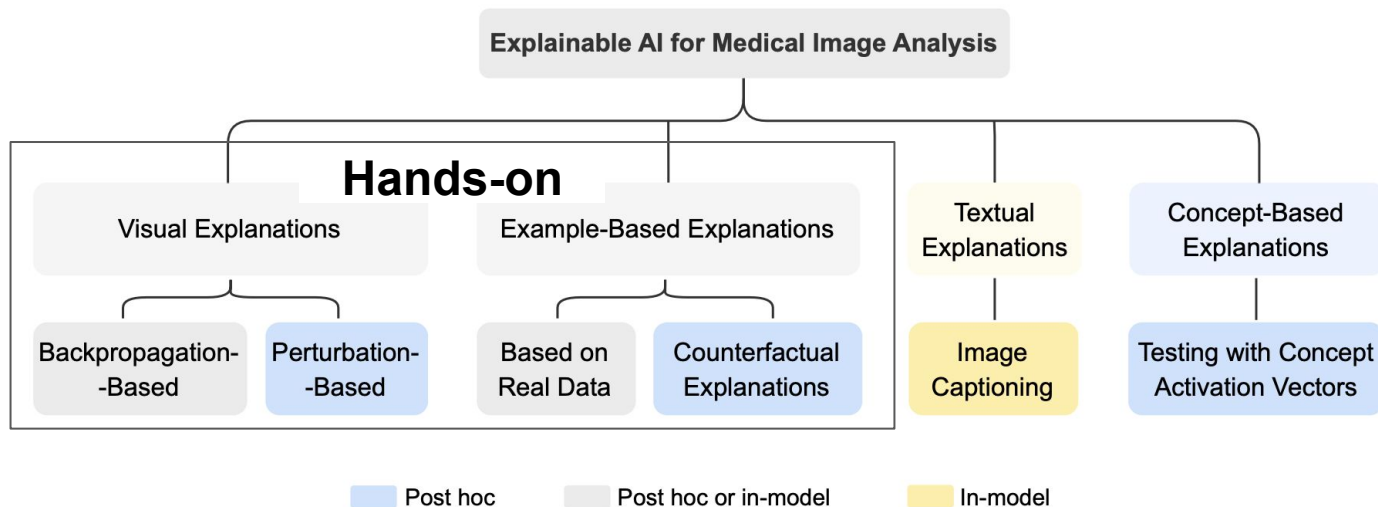


Example-based Explanations

- What are the **potential challenges/risks** of using **real data** to provide **example-based explanations**?

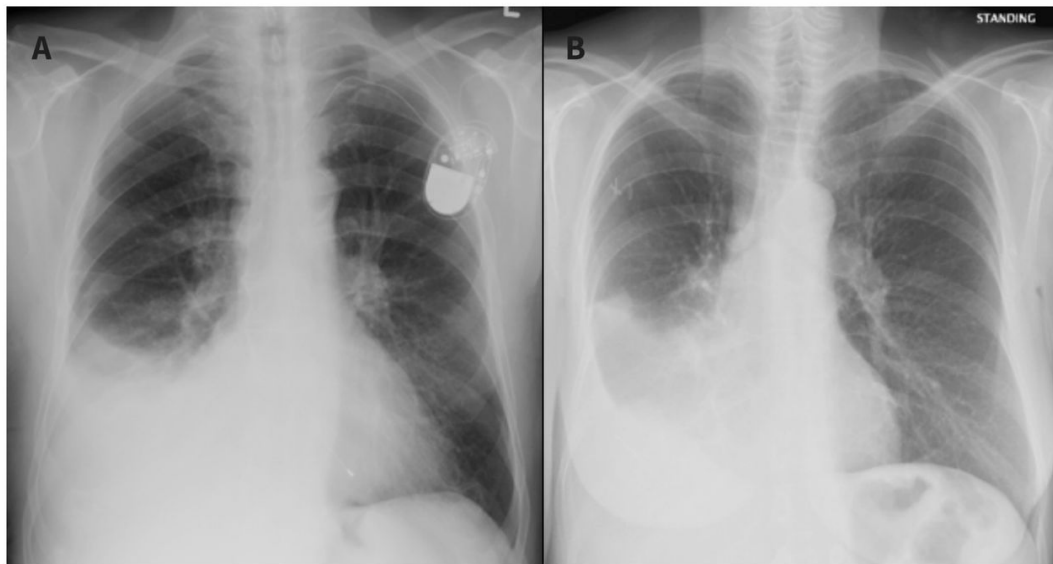


Explainable ML (for Medical Image Analysis)



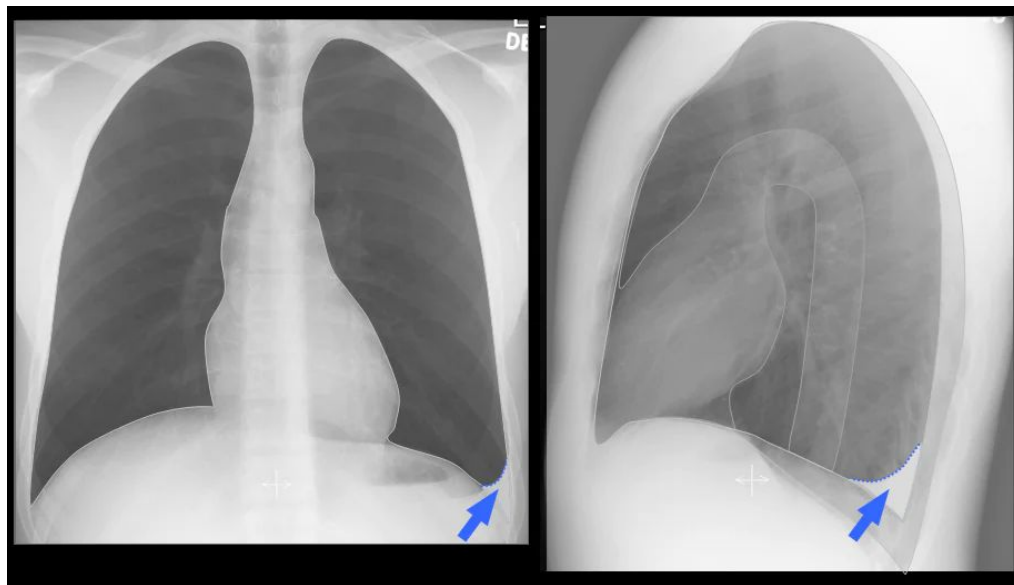
Hands-on: Explainable AI in a Radiology Scenario

- Goal is to **detect** and **explain pleural effusion** (saliency maps and example-based explanations)

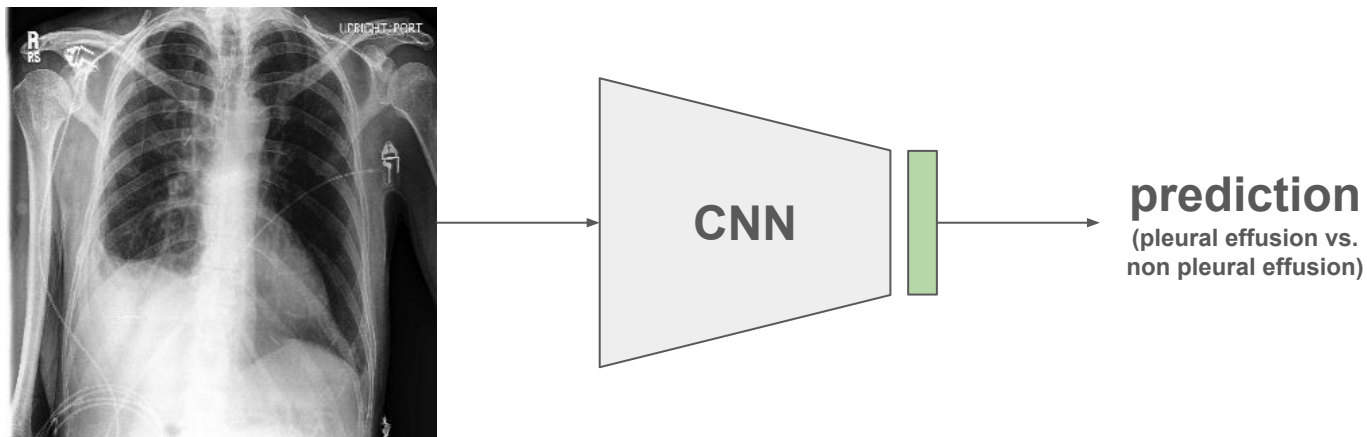


Hands-on: Explainable AI in a Radiology Scenario

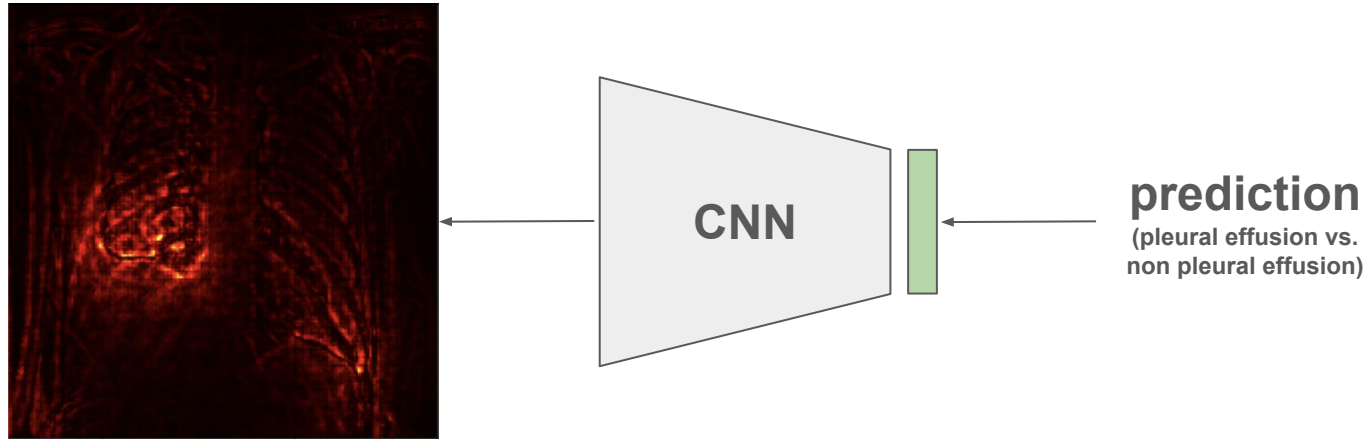
- **Pleural effusions** are abnormal **fluid accumulations** in the pleural space
- occur when pathological processes exceed the pleura's ability to reabsorb fluid



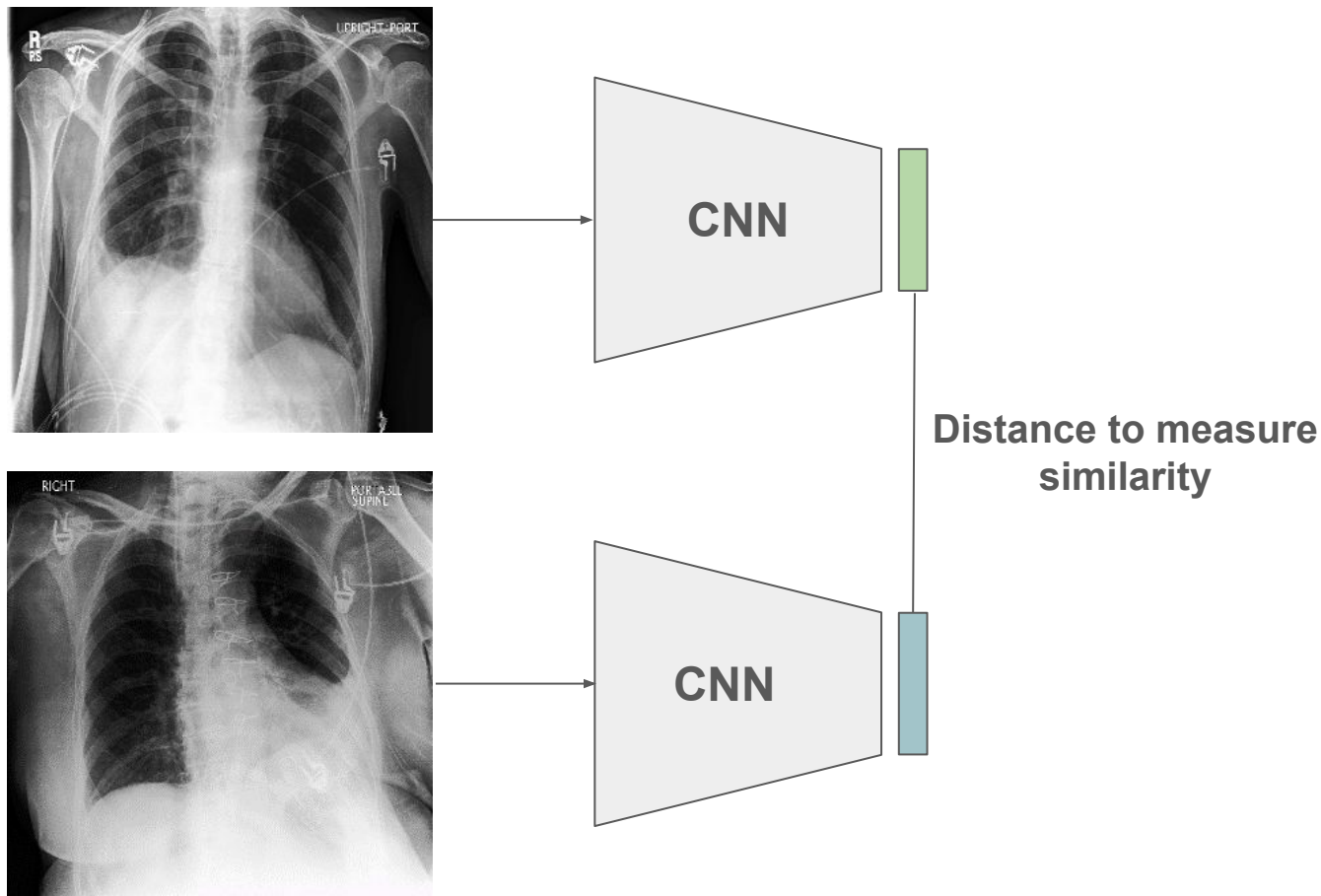
Hands-on: Explainable AI in a Radiology Scenario



Hands-on: Explainable AI in a Radiology Scenario



Hands-on: Explainable AI in a Radiology Scenario



Hands-on: Explainable AI in a Radiology Scenario

- Questions?

ProtoPNet

AI Model – Explainability Method		
Generation (method)	Ante-hoc/Post-hoc	
	Model-specific/Model-agnostic	
	Scope (Local/global)	



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2 Enter the event code in the top banner

Event code
QQRXSH

Enable answers by SMS

CONCEPT-BASED EXPLAINABLE AI

A LECTURE

Concept Bottleneck Models

Pang Wei Koh^{*1} Thao Nguyen^{*12} Yew Siang Tang^{*1}
Stephen Mussmann¹ Emma Pierson¹ Been Kim² Percy Liang¹

Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV)

Been Kim Martin Wattenberg Justin Gilmer Carrie Cai James Wexler
Fernanda Viegas Rory Sayres

Concept-based Explanations

Concept-based Explainable AI

- What is concept-based Explainable AI?
 - **AI decisions explainable** by associating model predictions with **semantic concepts**

Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV)

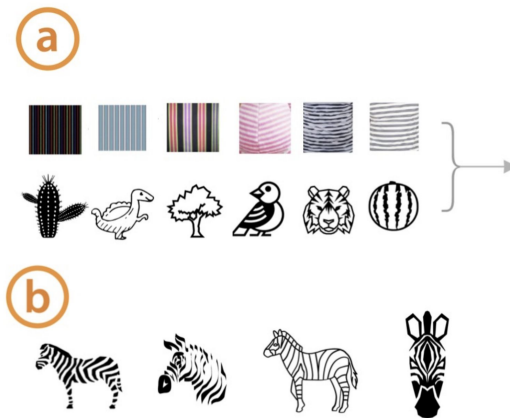
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Concept Bottleneck Models

Pang Wei Koh^{*1} Thao Nguyen^{*1,2} Yew Siang Tang^{*1}
Stephen Mussmann¹ Emma Pierson¹ Been Kim² Percy Liang¹

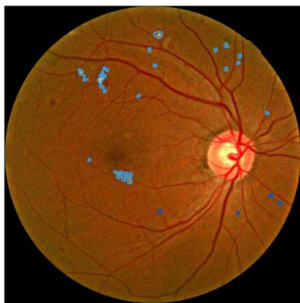
Quantitative Testing with CAVs

- Why are they relevant?
 - traditional **feature-attribution** methods are low-level and **hard to interpret**
 - **many** real-world **applications** require insights tied to **human-understandable concepts**

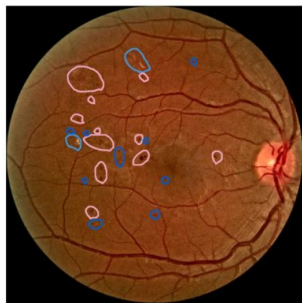


Quantitative Testing with CAVs

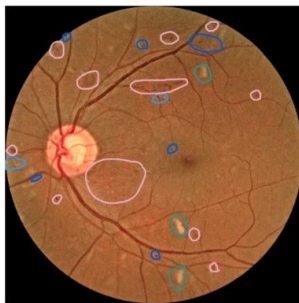
- How important is a concept to a model's prediction?
 - role of **striped pattern** in identifying zebras
 - how **hemorrhages** contribute to **diabetic retinopathy**



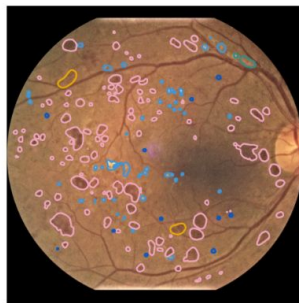
(a) DR level 0



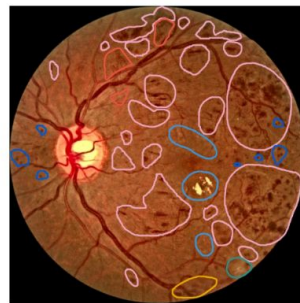
(b) DR level 1



(c) DR level 2



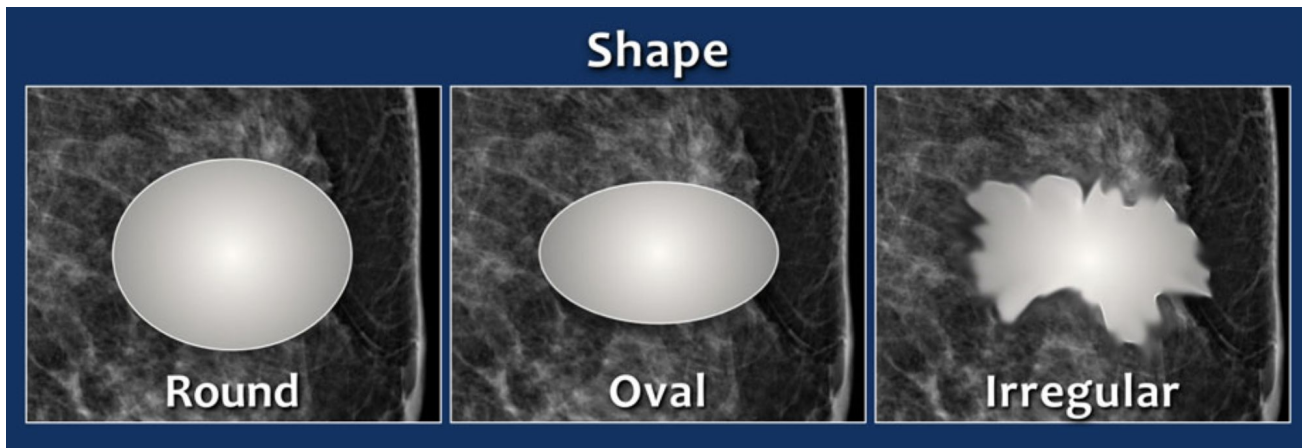
(d) DR level 3



(e) DR level 4

Quantitative Testing with CAVs

- What is a concept?
 - a human-understandable, high-level abstraction (e.g., stripes, breast mass shape)
 - represented by a set of examples (images) demonstrating the concept



Quantitative Testing with CAVs

- What are Concept Activation Vectors (CAVs)
 - **vectors** in the **direction** of the values (**activations**) of that concept's set of examples

a

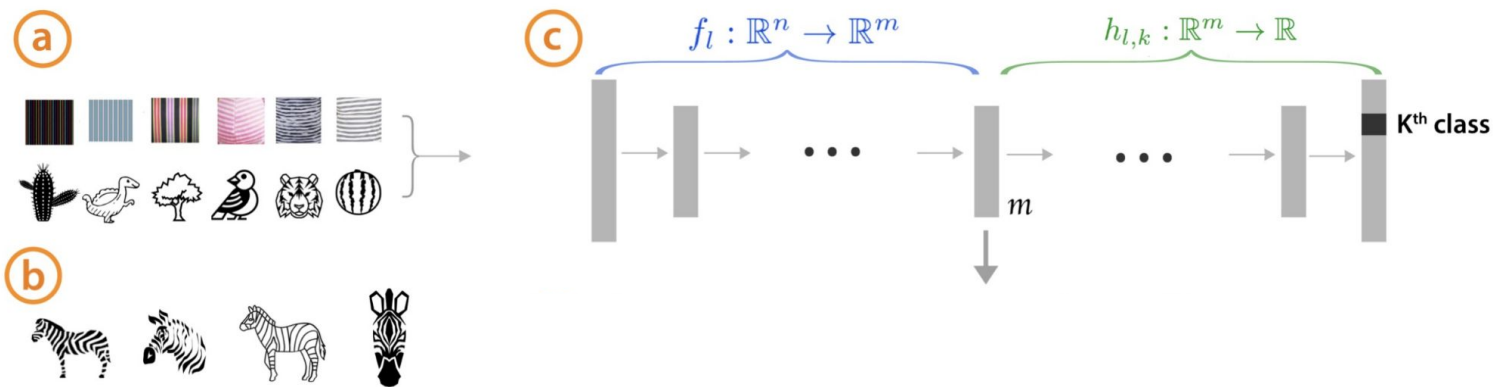


b



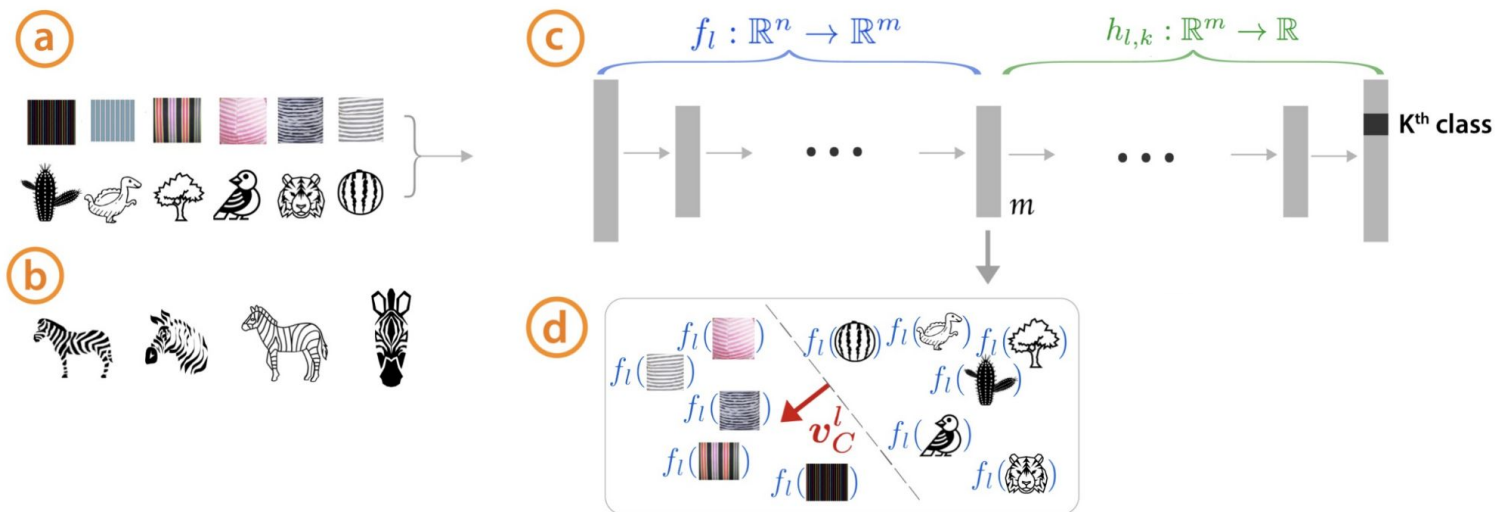
Quantitative Testing with CAVs

- What are Concept Activation Vectors (CAVs)
 - **vectors** in the **direction** of the values (**activations**) of that concept's set of examples



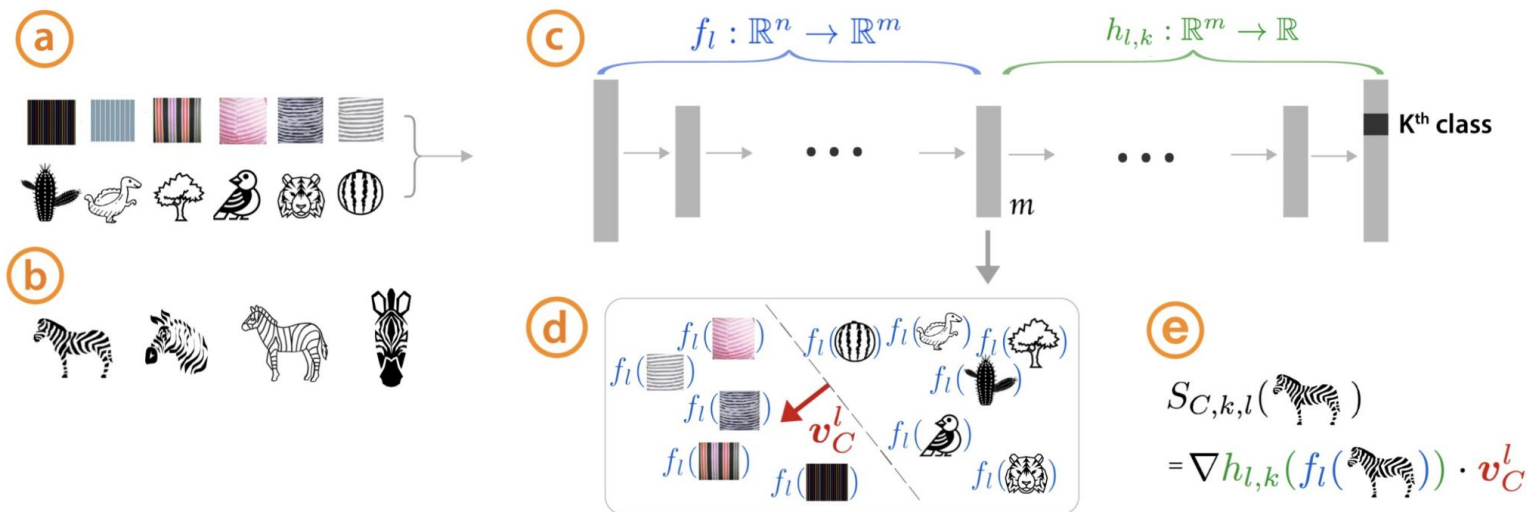
Quantitative Testing with CAVs

- What are Concept Activation Vectors (CAVs)
 - **vectors** in the **direction** of the values (**activations**) of that concept's set of examples



Quantitative Testing with CAVs

- What are Concept Activation Vectors (CAVs)
 - **vectors** in the **direction** of the values (**activations**) of that concept's set of examples



Quantitative Testing with CAVs

- How to measure concept importance?
 - **Saliency maps** use the **gradients of logit** values with respect to the individual **input features**

$$\frac{\partial h_k(\mathbf{x})}{\partial x_{a,b}}$$

- **Sensitivity** of ML **predictions** to changes in inputs **towards the direction** of a **concept**

$$\begin{aligned} S_{C,k,l}(\mathbf{x}) &= \lim_{\epsilon \rightarrow 0} \frac{h_{l,k}(f_l(\mathbf{x}) + \epsilon \mathbf{v}_C^l) - h_{l,k}(f_l(\mathbf{x}))}{\epsilon} \\ &= \nabla h_{l,k}(f_l(\mathbf{x})) \cdot \mathbf{v}_C^l, \end{aligned} \quad (1)$$

Quantitative Testing with CAVs

- Compute the TCAV score
 - the **TCAV score** for class k and concept C at layer l is the fraction of test inputs where the directional derivative is positive

$$\text{TCAV}_{Q_C,k,l} = \frac{|\{\mathbf{x} \in X_k : S_{C,k,l}(\mathbf{x}) > 0\}|}{|X_k|} \quad (2)$$

- a **high TCAV score** suggests that a **model** strongly **relies** on the **concept** when making **predictions** for a specific class

Quantitative Testing with CAVs

- **Advantages**

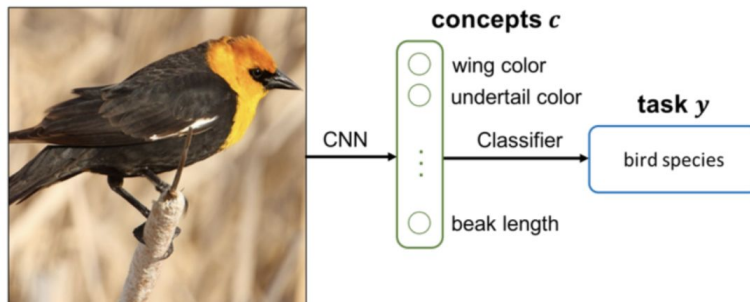
- **human-interpretable** (model predictions explained with clinically relevant concepts)
- **model-agnostic** (works with any neural network)
- **no** need for model **retraining** (based on activations from an existing model)

- **Disadvantages**

- **only global explanations** (cannot localize where a concept appears)
- **requires curated concept sets** (human effort is needed to collect labelled examples)

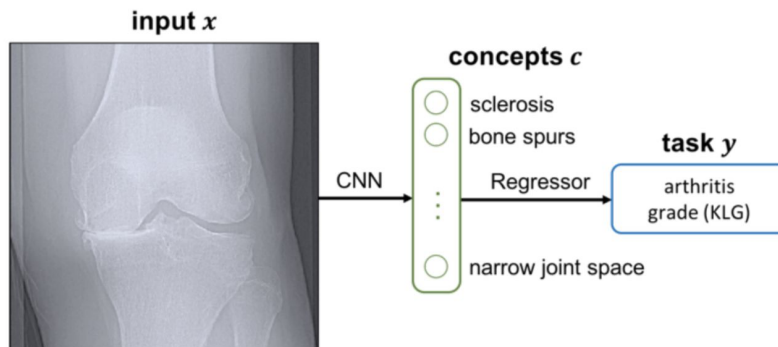
Concept Bottleneck Model

- What are CBMs?
 - **design neural networks** to pass **information** through a **bottleneck** of **interpretable concepts** before making a prediction
 - **unlike TCAVs**, **ensure** that the **model** explicitly uses human-interpretable concepts



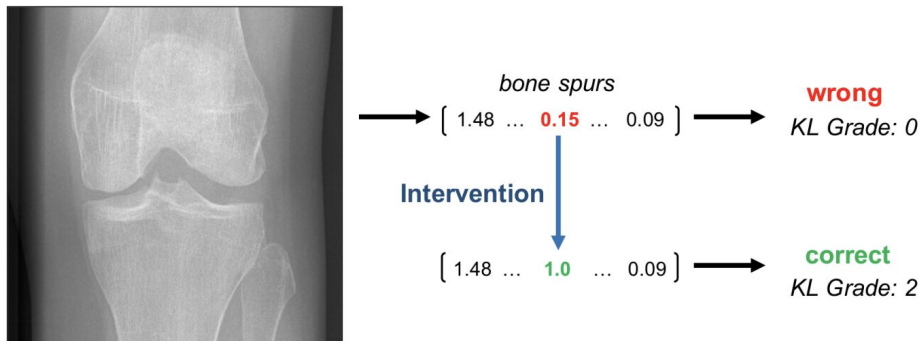
Concept Bottleneck Model

- What are CBMs?
 - predicts human-defined **concepts based** on raw **input x**
 - uses **concepts to** make a final **prediction**



Concept Bottleneck Model

- What are CBMs?
 - loss for predicting concepts c
 - loss for predicting the final task output y
- Users can **intervene** and **correct** concept misclassifications



Concept Bottleneck Model

- **Advantages**

- human-interpretable (model predictions explained with relevant concepts)
- enforce concept-based reasoning during training (less reliance on spurious correlations)

- **Disadvantages**

- concept annotation cost (require datasets labelled with concepts)
- requires redesigning architectures

Concept Bottleneck Model

- Why do CBMs face ambiguity issues?

Class: Green Jay

Concepts:

forehead color: blue

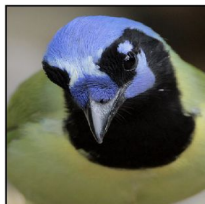
throat color: black

belly color: yellow

tail pattern: solid



Diverse visual contexts



ambiguity in tail



ambiguity in belly



ambiguity in color



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Concept Bottleneck Model

- **Uncertainty issue** with CBMs
 - **CBMs** treat **concept predictions** as **deterministic** values
 - if a **concept** is **uncertain**, a CBM still commits to a fixed prediction

Class: Green Jay

Concepts:

forehead color: blue
throat color: black
belly color: yellow
tail pattern: solid



Diverse visual contexts



ambiguity in tail



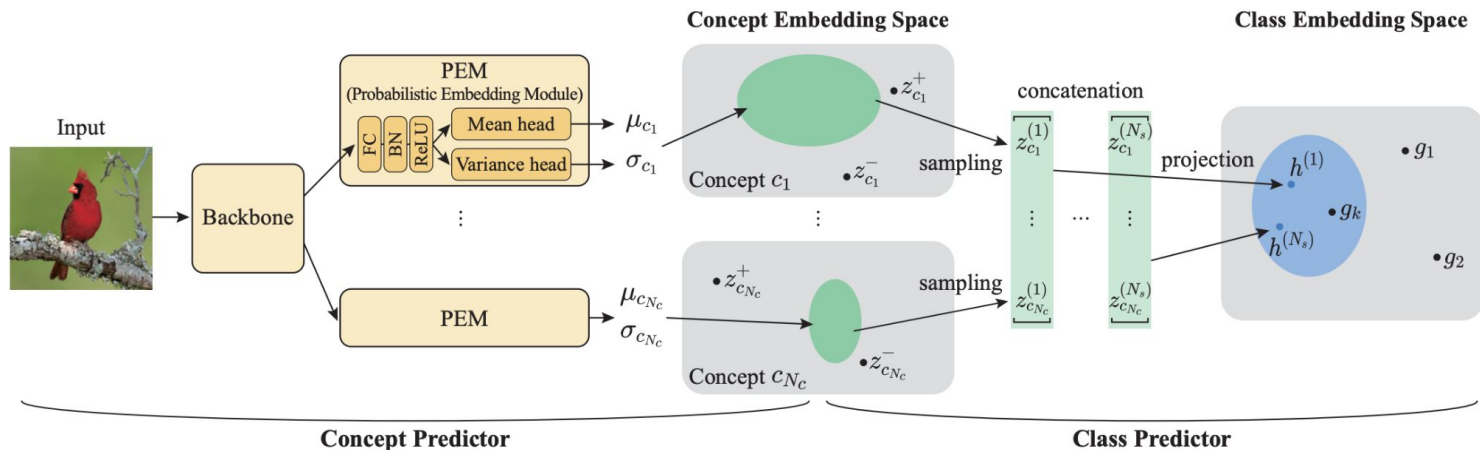
ambiguity in belly



ambiguity in color

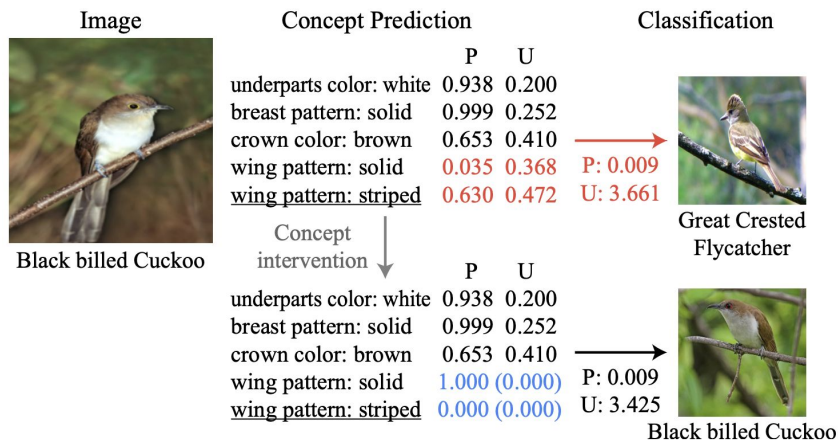
Probabilistic Concept Bottleneck Model

- What are Probabilistic CBMs?
 - model **concepts** as **probability distributions** instead of fixed values
 - helps represent **uncertainty** in concept presence



Probabilistic Concept Bottleneck Model

- What are Probabilistic CBMs?
 - **class embedding** is **derived** from **probabilistic concept embedding**
 - thus, able to provide **uncertainty** in class **prediction**



Quantitative Testing with CAVs

AI Model – Explainability Method		
Generation (method)	Ante-hoc/Post-hoc	
	Model-specific/Model-agnostic	
	Scope (Local/global)	





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Concept Bottleneck Model

AI Model – Explainability Method		
Generation (method)	Ante-hoc/Post-hoc	
	Model-specific/Model-agnostic	
	Scope (Local/global)	





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Is Attention All You Need?

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

Alexey Dosovitskiy^{*,†}, Lucas Beyer^{*}, Alexander Kolesnikov^{*}, Dirk Weissenborn^{*},
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Attention Is All You Need

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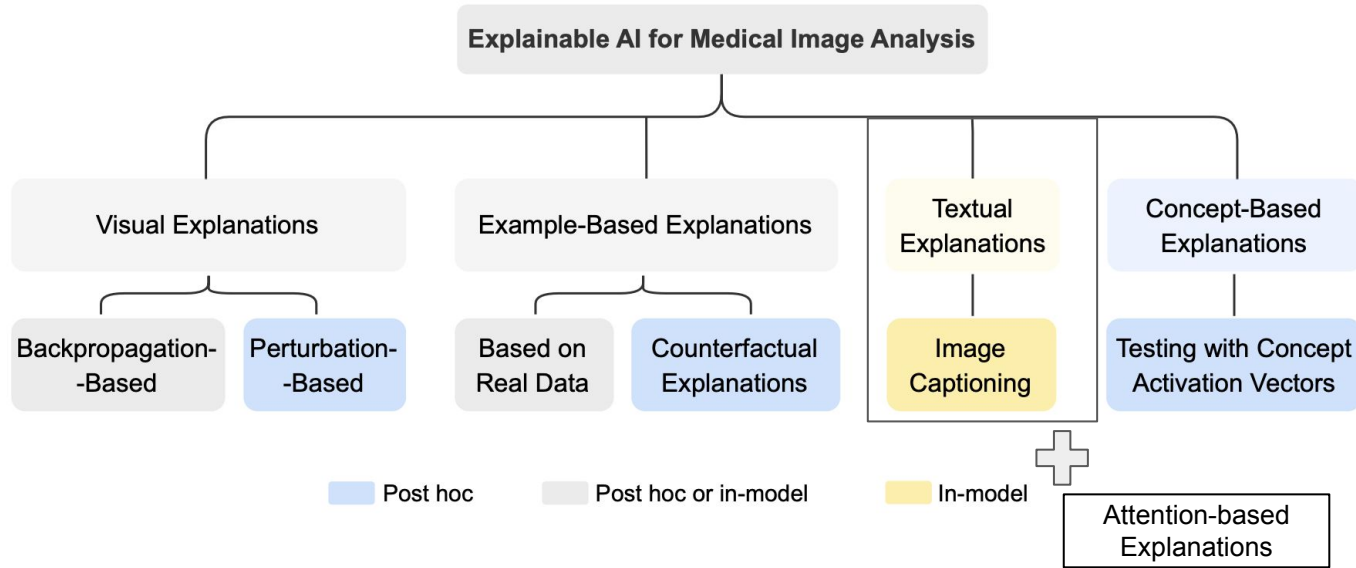
Aidan N. Gomez^{*} [†]
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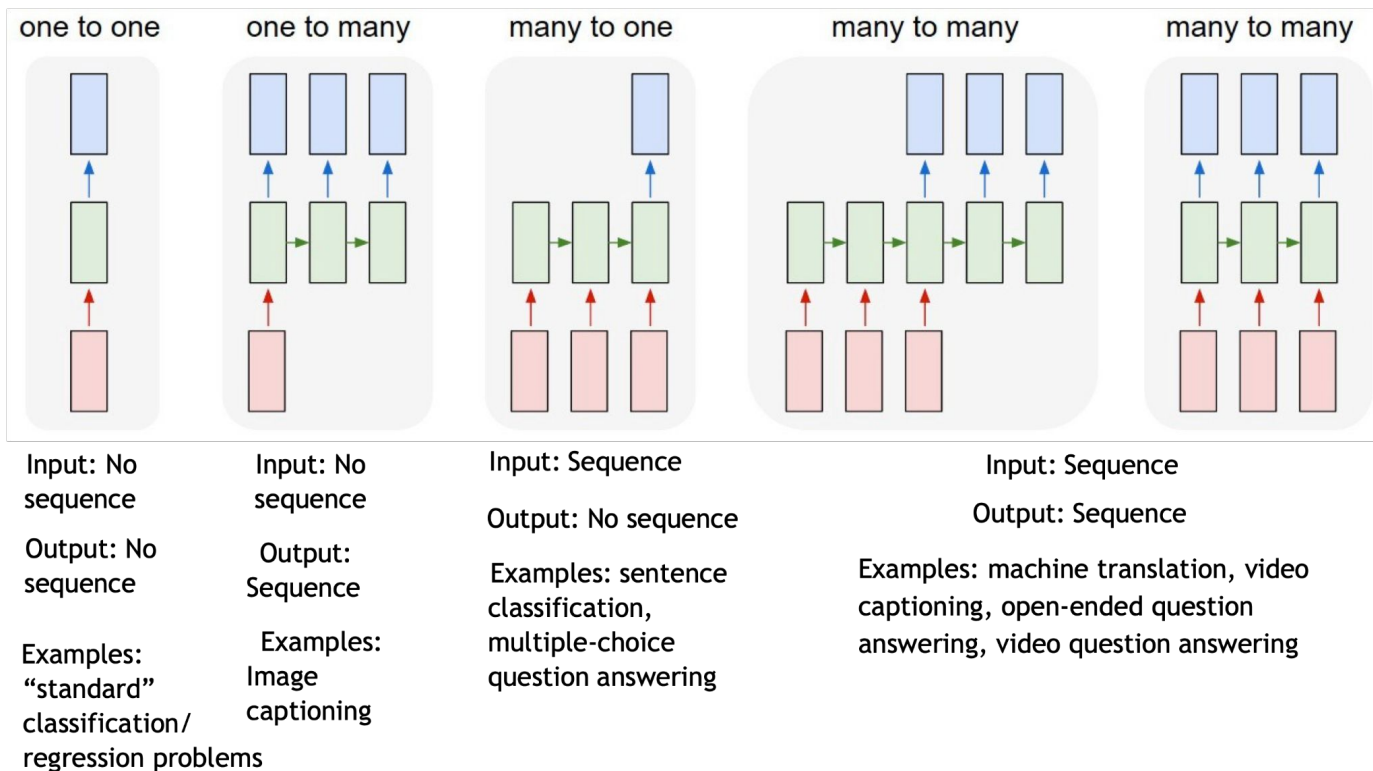
Illia Polosukhin^{*} [†]
illia.polosukhin@gmail.com

Attention-based Explanations and Image Captioning

Explainable ML (for Medical Image Analysis)

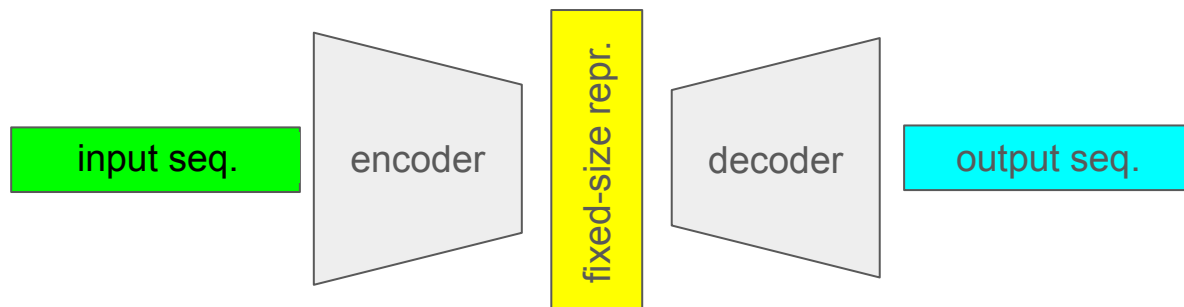


Sequence modeling



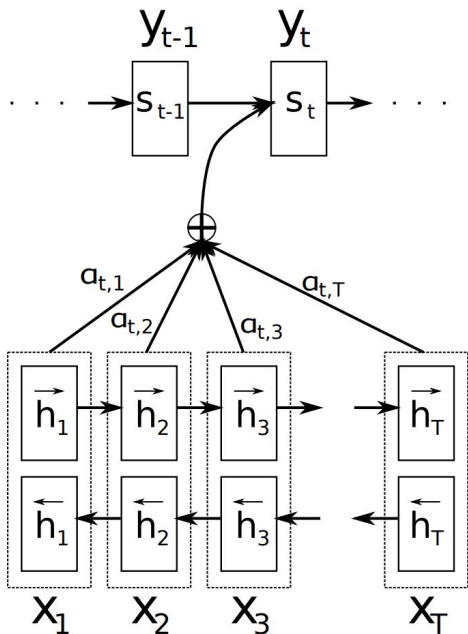
On the origins of attention - seq2seq models

- Standard **seq2seq models encode** the entire **input** sequence into a single **fixed-length vector**
 - model is forced to **compress** all **information** into **one representation**
 - **long-range** dependencies are **hard** to maintain



On the origins of attention - seq2seq models

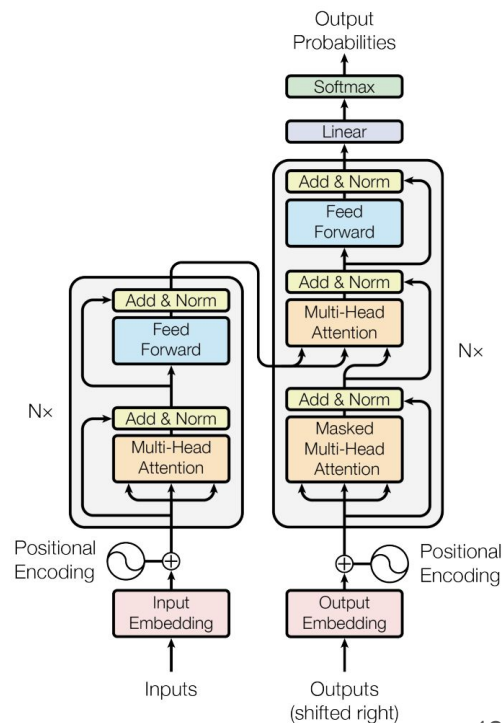
- Key components
 - contextual hidden states (encoded from the input sequence)
 - decoder assigns **attention weights** (alignment) to different states
 - attention-weighted vectors to generate the output sequence



Attention is all you need

- **Transformer**

- **every token attends to all other tokens** simultaneously
- learns multiple attention patterns at different levels of abstraction
- **positional encodings** preserve word order



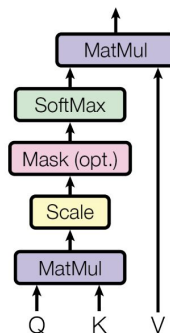
Attention is all you need

- **Transformer**

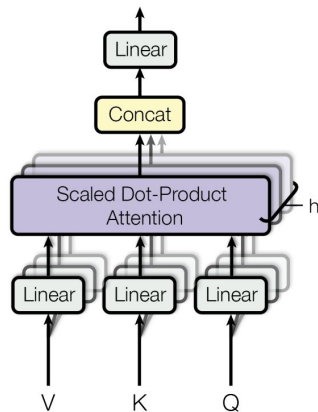
- **dot product similarity** between queries (Q) and keys (K)
- softmax to get **attention weights**
- retrieves the weighted sum of V

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Scaled Dot-Product Attention

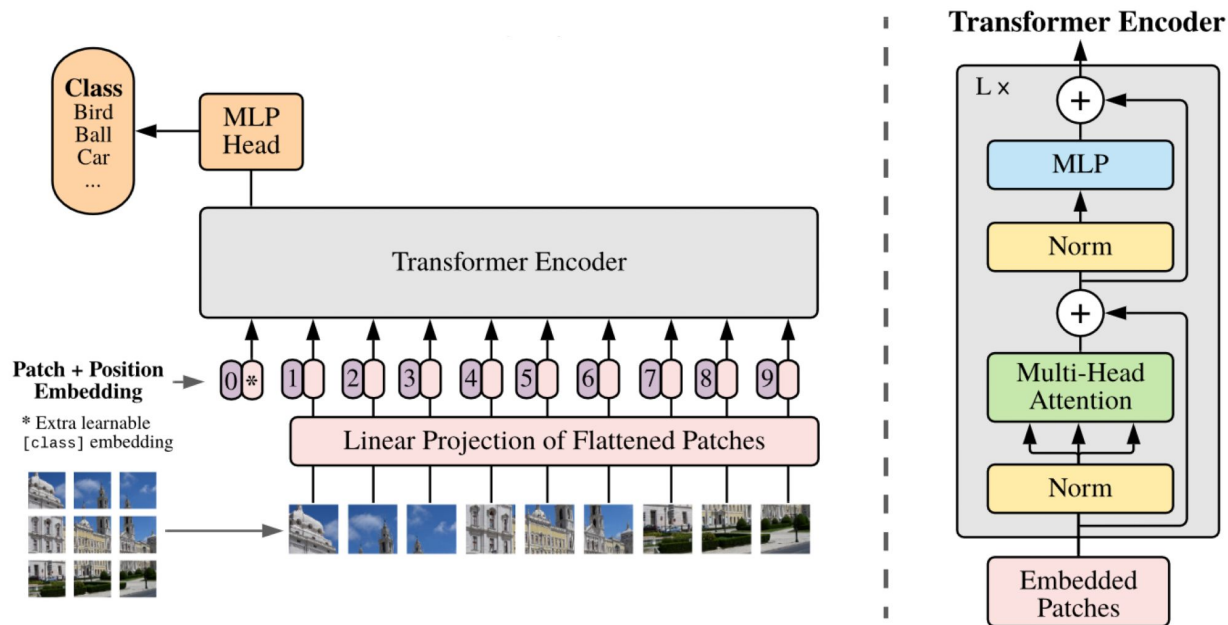


Multi-Head Attention



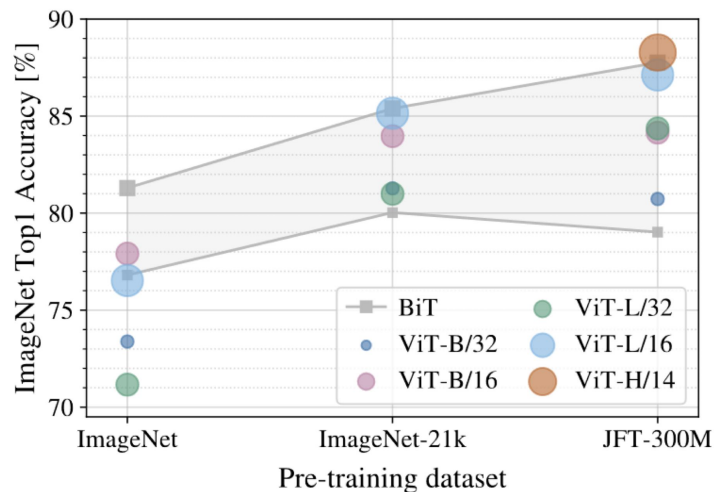
Vision Transformer

- ViT divides an **image** into **patches** and processes them as a sequence using self-attention



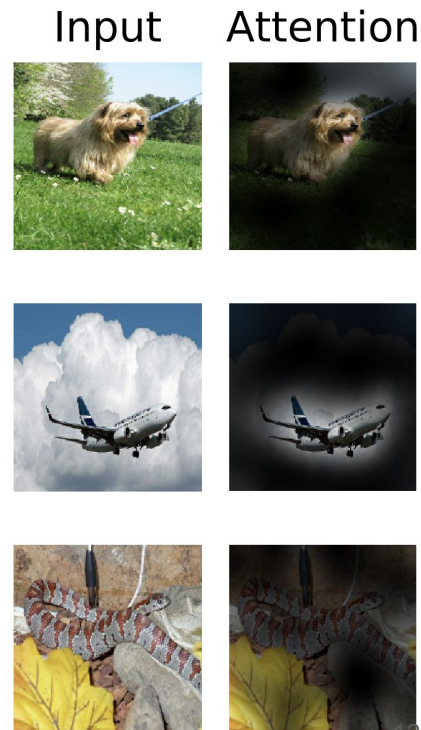
Vision Transformer

- **ViT** divides an **image into patches** and processes them as a sequence using self-attention
 - lack of CNN-like inductive biases (e.g., translation equivariance)
 - poor generalization on limited data
 - large-scale training overcomes these limitations



Vision Transformer

- ViT divides an **image into patches** and processes them as a sequence using self-attention
 - ViTs use **self-attention** to process the **entire image** globally
 - **attention weights** represent the **importance** of each image **patch**
 - by **visualizing attention maps**, we can **interpret** model



Vision Transformer

- **Limitations of attention as explanation**

- just because a model attends to a region **does not mean that region “caused” the decision**
- **different heads** capture **different things**, hard to say which attention map is most meaningful
- a model can reweight attention without changing outputs (**not robust** explanations)

Vision Transformer

- If we don't trust attention, what alternative explainability methods could we use?



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Image captioning

- Textual explanations employ the use of natural language to describe the model's behavior
 - process of describing visual content using **natural language**
 - combines a **visual model** with a **language model** to generate meaningful sentences
 - both models are trained in a **shared embedding space**, linking vision and language

Image captioning

- Since the two models are trained in a **shared embedding** space
 - **image captions** act as **explanations**
 - can speed up medical report generation, reducing workload and costs

Image captioning

- Early image captioning methods
 - used **Recurrent Neural Networks** (RNNs), particularly Long Short-Term Memory (LSTM)
 - single-layer LSTM with **vision model's encoding** as the **initial hidden state**
 - can be enhanced with additive attention, generating a context vector

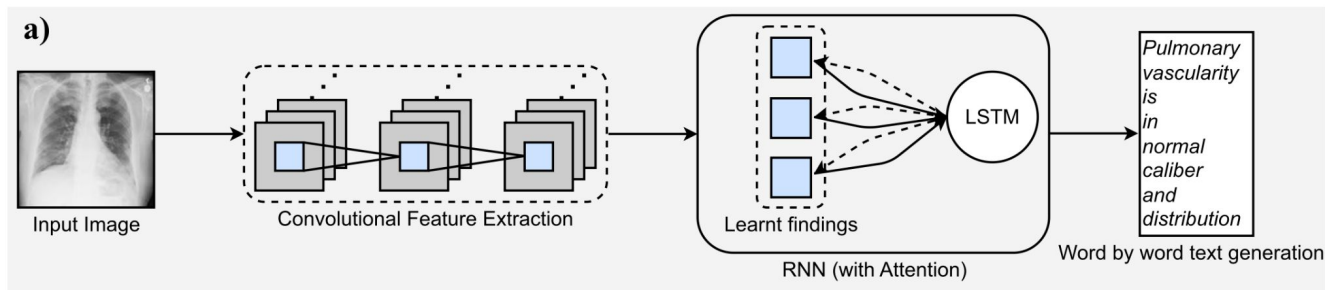


Image captioning

- Limitations of CNN-RNN approach
 - RNNs extract only **top-down visual features** (potential lack of coherence)
 - Transformers improve by better **context retention** and **cross-attention** mechanisms

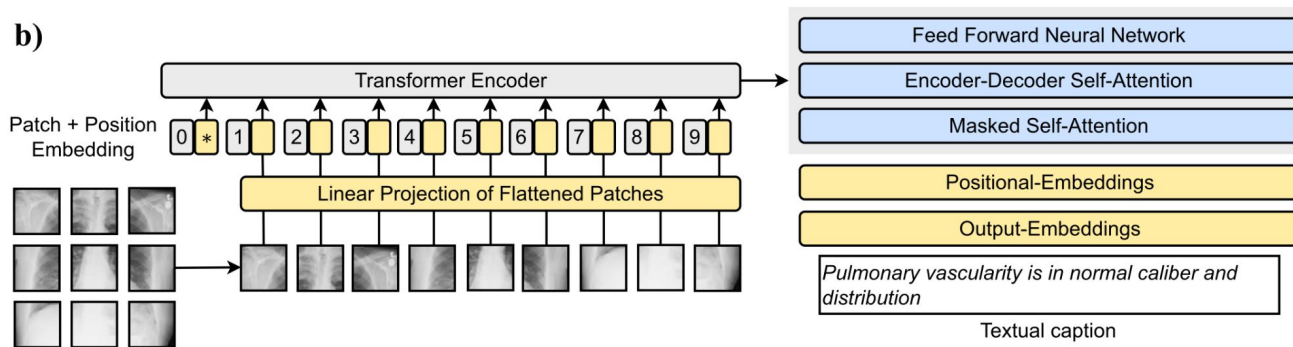


Image captioning

- Transformer-based captioning methods
 - **masked self-attention**: model generates words sequentially
 - **cross-attention**: **words** acting as **queries**, **visual features** as **keys** and **values**
 - final step is always to **predict the next word**

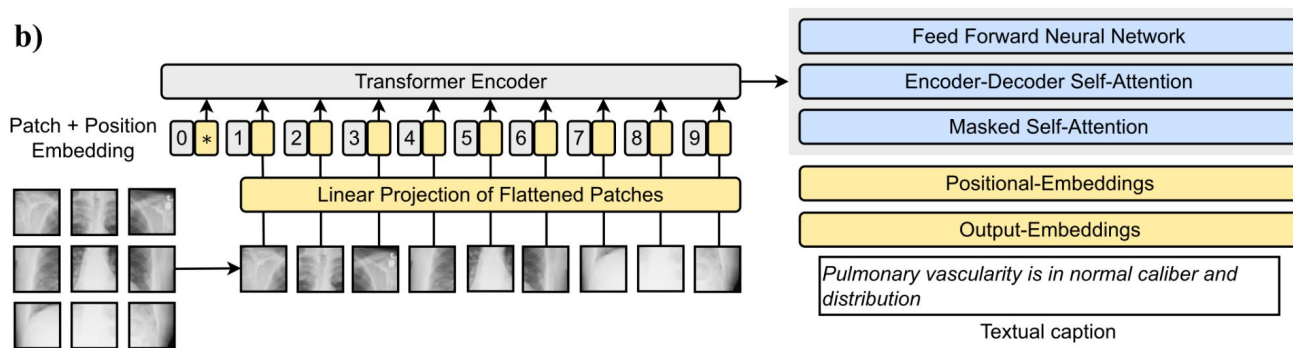


Image captioning

- Whole-slide image captioning
 - large-scale knowledge of pretrained vision transformers can be tailored for downstream tasks
 - weakly-supervised **transformer-based model** for **classification** and pathology **report generation**
 - multicentric multilingual dataset of colon polyps and biopsies

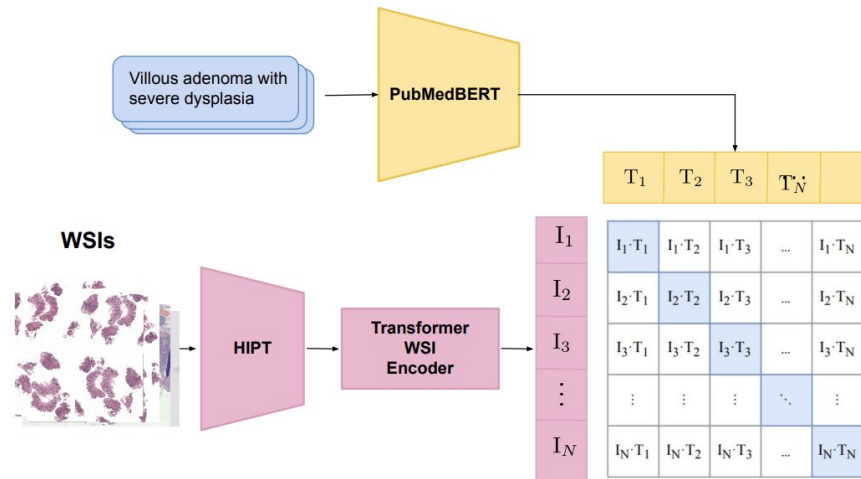


Image captioning

- Whole-slide image captioning
 - HIPT model to **encode WSIs** and PubmedBERT to **encode medical text**
 - 1st stage: contrastive WSI-caption pre-training
 - 2nd stage: decoder blocks are conditioned on the WSI embeddings

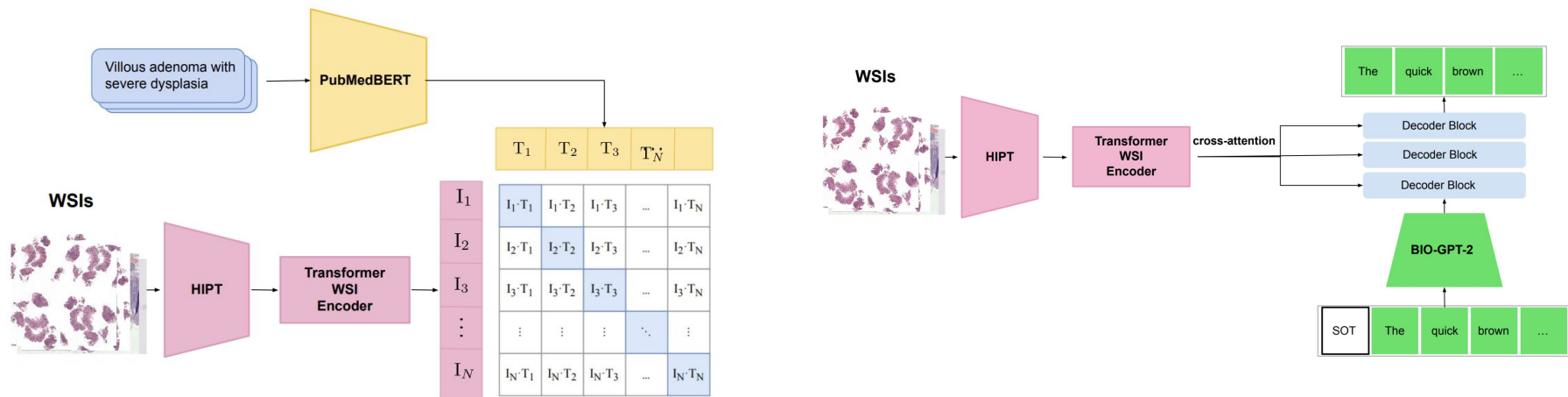


Image captioning

- Potential **limitations** of image captioning





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Image captioning

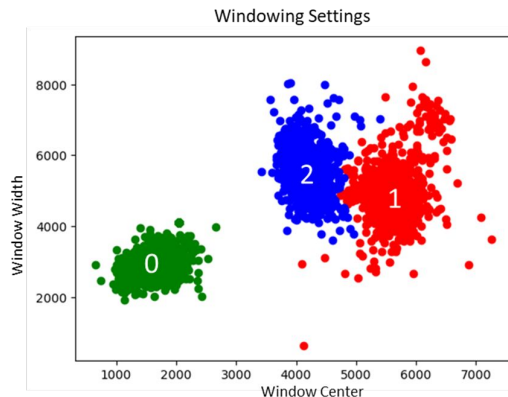
- Potential **limitations** of **image captioning**
 - **may not** accurately **reflect** the actual **decision-making process** of the model
 - generated text might **align** with **human expectations rather** than model **reasoning**
 - leading to **descriptive but not explanatory** outputs
 - may **introduce biases** that distort explanations

Beyond conventional explainable AI



- **Attention-based regularisation**

- AI models are **sensitive** to **confounders**
- **Image acquisition** and **preprocessing** vary
- **Windowing** changes **contrast** and **brightness**

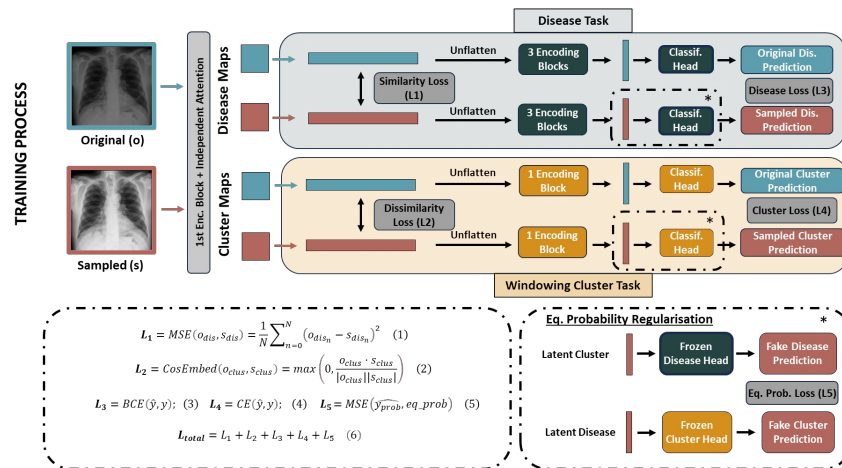


Beyond conventional explainable AI



- **Attention-based regularisation**

- Diagnosis **invariant** to dataset/windowing
- Promote **similar disease attention maps**
- Promote **dissimilar windowing attention maps**
- **Latent representations** blind to opposite task



Beyond conventional explainable AI



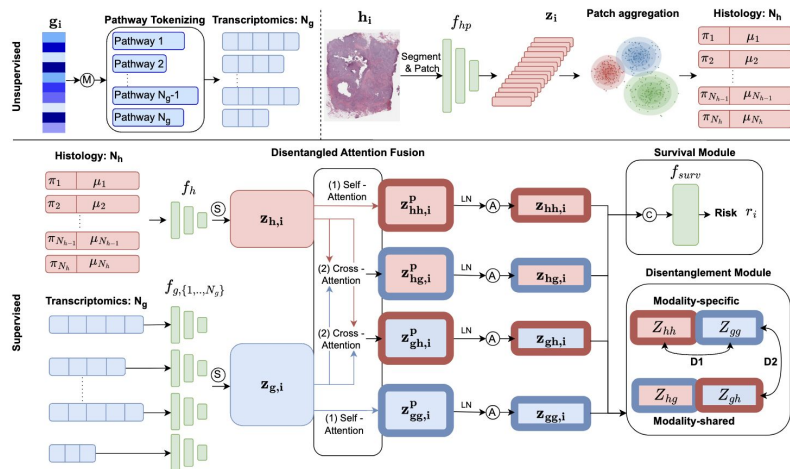
- Attention-based regularisation
 - proposed approach **outperforms baseline** across three out-of-distribution datasets
 - it does not compromise in-distribution performance

FOLD	Out-of-distribution testing					
	MIMIC-CXR-JPG		CheXpert		VinDr-CXR	
	Baseline	Proposal	Baseline	Proposal	Baseline	Proposal
1	72.77	77.91	75.95	75.17	61.07	67.77
2	75.83	77.39	79.50	76.21	65.23	70.43
3	76.47	80.18	75.41	81.46	61.46	71.17
4	77.58	79.84	78.46	78.86	60.69	68.78
5	77.58	75.87	75.41	74.08	68.28	67.90
Average	76.05 $\pm$ 1.98	78.24 $\pm$ 1.79	76.95 $\pm$ 1.91	77.16 $\pm$ 2.99	63.35 $\pm$ 3.30	69.21 $\pm$ 1.53

Beyond conventional explainable AI



- **Attention-based regularisation** (Disentangled Representation Learning)
 - **integrating** life science **data** from **diverse modalities** (omics and medical imaging)
 - **disentangled representation learning** can improve model interpretability
 - **transformer-based fusion** of the different **modalities**



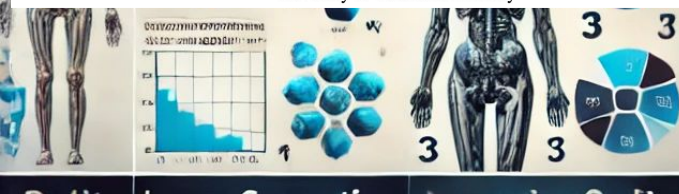


Sanity Checks for Saliency Maps

Julius Adebayo*, Justin Gilmer[‡], Michael Muelly[‡], Ian Goodfellow[‡], Moritz Hardt^{†‡}, Been Kim[‡]
 juliusad@mit.edu, {gilmer,muelly,goodfellow,mrtz,beenkim}@google.com

[‡]Google Brain

[†]University of California Berkeley



Saliency Map Explanations

1.

Quantus: An Explainable AI Toolkit for Responsible Evaluation of Neural Network Explanations and Beyond

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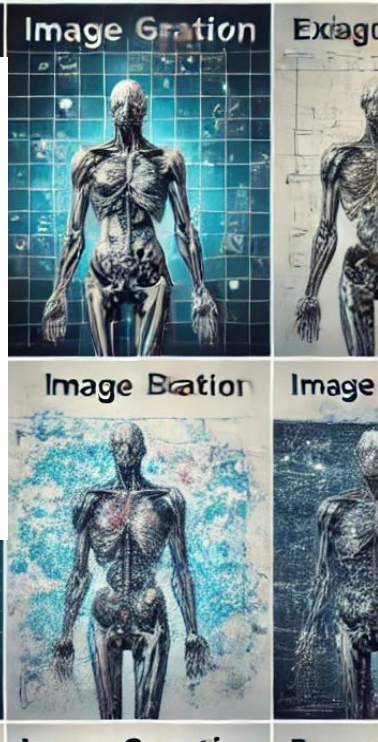
² Department of Electrical Engineering and Computer Science, TU Berlin, 10587 Berlin, Germany

³ Department of Artificial Intelligence, Fraunhofer Heinrich-Hertz-Institute, 10587 Berlin, Germany

⁴ Department of Computer Science, University of Potsdam, 14476 Potsdam, Germany

⁵ BIFOLD – Berlin Institute for the Foundations of Learning and Data, 10587 Berlin, Germany

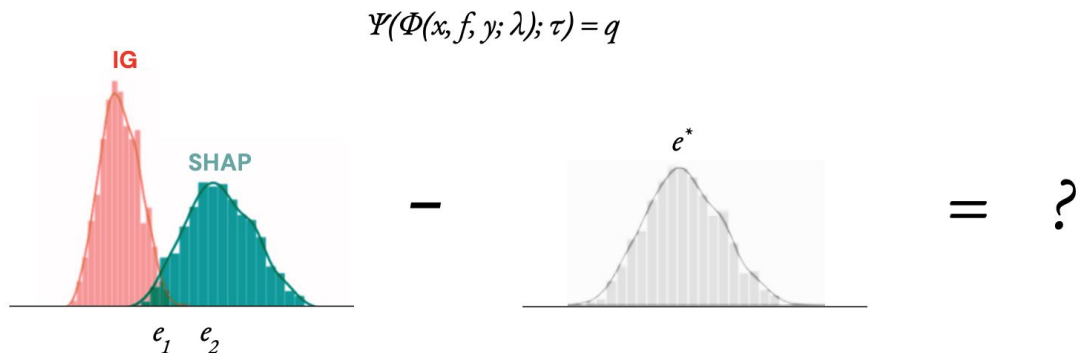
[†] corresponding authors



Assessment of Explanations

The Challenge of Explanation Evaluation

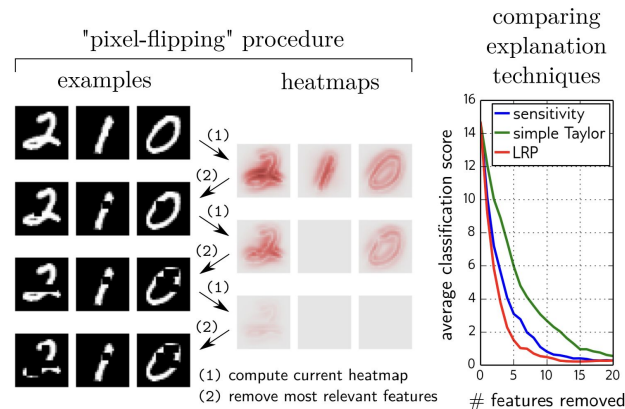
- While selecting an explanation method for a user, we face evaluation difficulties
 - without ground-truth explanation labels, we can't compute explanation error
 - the lack of ground-truth explanation labels e^* forms the evaluation problem in explainable AI



Estimating Explanation Quality - Faithfulness

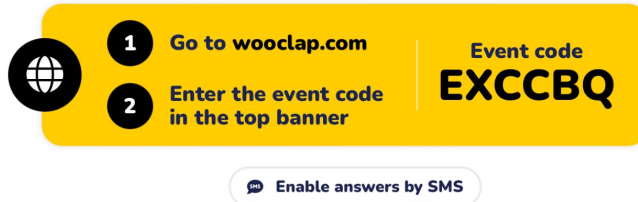
- **Pixel-flipping**: captures the model degradation while perturbing pixels in the input based on the explanation output
 - removes pixels gradually, step by step, from an image
 - at each step i , a fraction of pixels is removed, and the model output is recorded
 - at the next step $i + 1$, more pixels are removed, and the new model's output is recorded

$$\Psi_{\text{PF}} = \sum_{i=1}^n (\hat{y}_i + \hat{y}_{i+1}) \cdot \frac{p_{i+1} - p_i}{2}$$



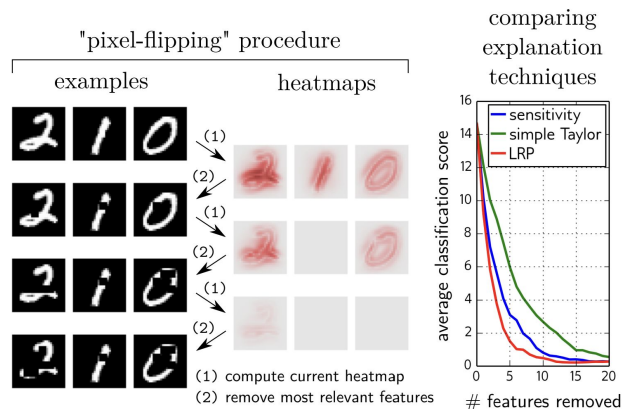
Estimating Explanation Quality - Faithfulness

- What are some potential issues with using **pixel-flipping** as an explainability metric?



Estimating Explanation Quality - Faithfulness

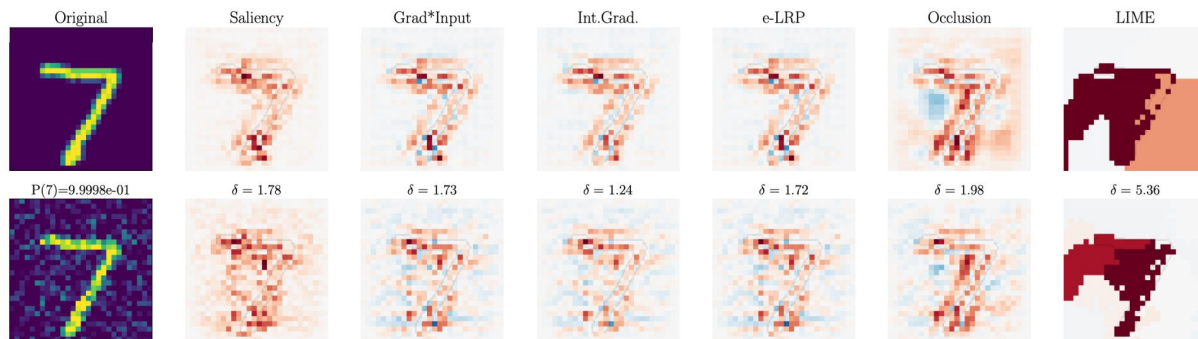
- **Pixel-flipping**: potential issues
 - dependency on the perturbation strategy (might cause distribution shifts)
 - lack of standardization (e.g. number of pixels removed per step)
 - sensitivity to model architecture (e.g., difference between CNNs and Transformers)



Estimating Explanation Quality - Robustness

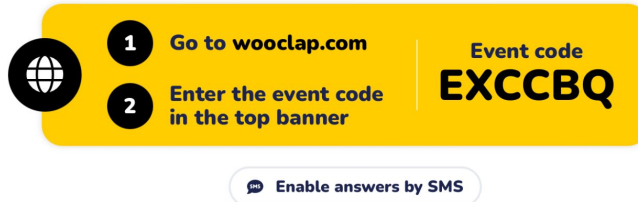
- **Local Lipschitz Estimate:** measures how much the explanation changes for the input under slight perturbation
 - a small perturbation is added to the input ($x + \delta$)
 - ϵ defines the range of allowed perturbations
 - a robust explanation should not change drastically when the input is slightly modified

$$\Psi_{\text{LLE}} = \max_{x+\delta \in \mathcal{N}_\epsilon(x) \leq \epsilon} \frac{\|\Phi(x, f, \hat{y}; \lambda) - \Phi(x + \delta, f, \hat{y}; \lambda)\|_2}{\|x - (x + \delta)\|_2}$$



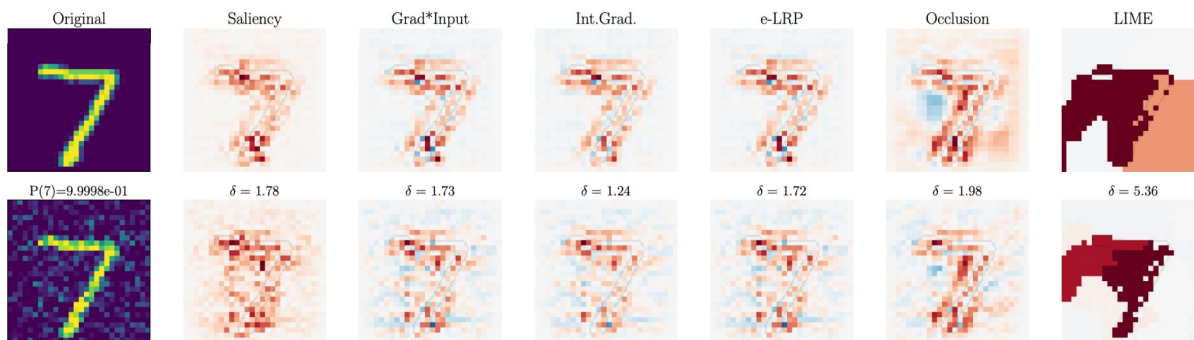
Estimating Explanation Quality - Robustness

- What are some potential issues with using **Local Lipschitz Estimate** as an explainability metric?



Estimating Explanation Quality - Robustness

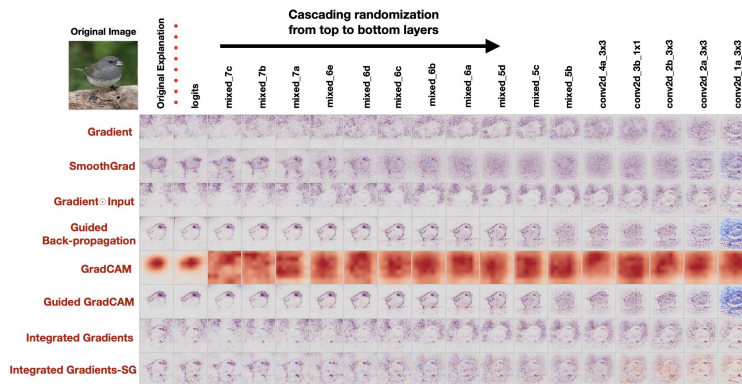
- **Local Lipschitz Estimate:** potential issues
 - explanation to model misalignment
 - an explanation can be stable but not faithful
 - an explanation should mirror model behavior, not just be inherently stable



Estimating Explanation Quality - Randomisation

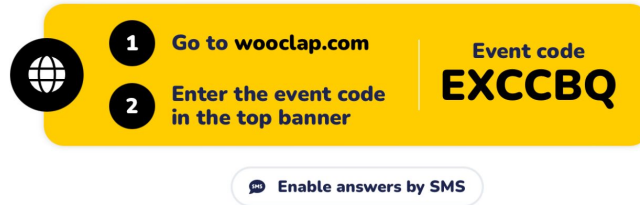
- **Model parameter randomisation test:** randomises the parameters of a single model layers top-down way, measuring the distance of the respective explanation to the original explanation
 - explanation from the trained model f
 - explanation from the randomized model
 - correlation function between the two explanations

$$\Psi_{\text{MPR}} = \frac{1}{V} \sum_{v=1}^V \text{corr}(\Phi^v(x, f, \hat{y}; \lambda), \Phi^v(x, \hat{f}, \hat{y}; \lambda))$$



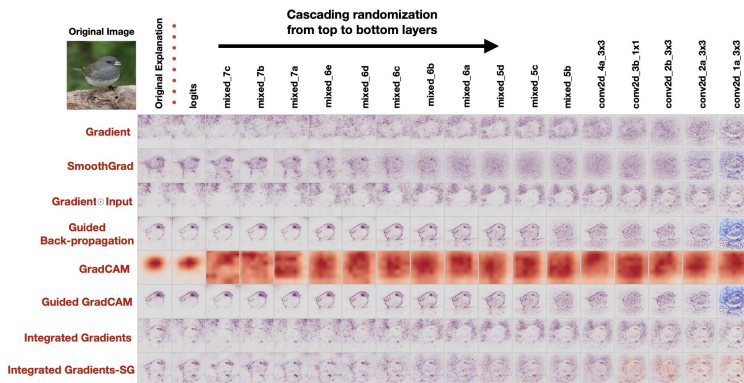
Estimating Explanation Quality - Randomisation

- What are some potential issues with using **model parameter randomization test** as an explainability metric?



Estimating Explanation Quality - Randomisation

- **Model parameter randomisation test:** potential issues
 - randomization order matters
 - structural changes in explanations might not be reflected in correlation-based scores
 - some saliency methods naturally smooth out variations



Estimating Explanation Quality - Localisation

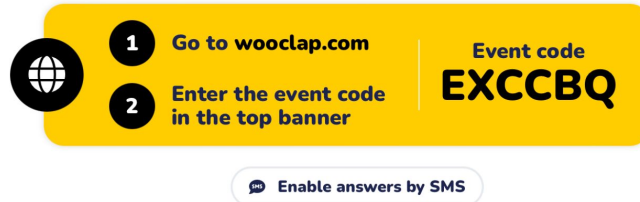
- **Relevance mass accuracy**: quantifies the fraction of the sum of the attribution that intersects with the ground truth mask over the full explanation sum
 - based on ground-truth importance mask
 - how much of the total relevance mass is assigned to the ground-truth important pixels
 - the total relevance mass assigned by the explanation method across all pixels

$$\Psi_{\text{RMA}} = \frac{\sum_{i=1}^D \Phi_i(x, f, \hat{y}; \lambda) \cdot s_{gt,i}}{\sum_{i=1}^D \Phi_i(x, f, \hat{y}; \lambda)}$$

What is the material of the large block? <i>metal</i>			GT Single Object
LRP [20]			0.94
Excitation Backprop [37]			0.82
IG [19]			0.99
Guided Backprop [22]			0.80

Estimating Explanation Quality - Localisation

- What are some potential issues with using **relevance mass accuracy** as an explainability metric?



Estimating Explanation Quality - Localisation

- **Relevance mass accuracy:** potential issues
 - segmentation masks defined by experts might not reflect the actual decision-making process
 - assumes clear distinction (binary masks) between relevant and irrelevant regions
 - larger ground-truth regions inherently produce higher scores due to greater overlap potential

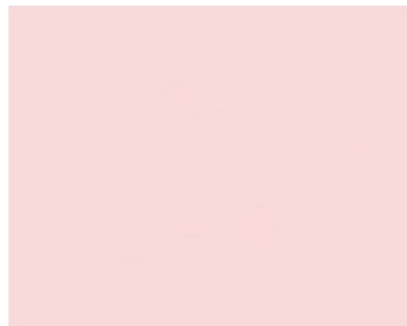
What is the material of the large block? <i>metal</i>			GT Single Object
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Estimating Explanation Quality - Complexity

- **Complexity:** computes the entropy of the fractional distribution of all features to the total magnitude of the attribution individually
 - how uniformly the explanation mass is distributed across features
 - entropy is maximized when importance is evenly distributed across all features
 - normalized probability representing how much mass of explanation is allocated to feature i

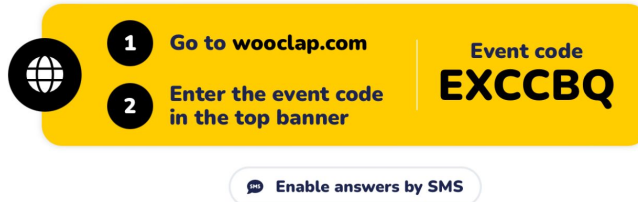
$$\Psi_{CO} = \mathbb{E}_i[-\ln(\mathbb{P}_\Phi)] = -\sum_{i=1}^D \mathbb{P}_\Phi(i) \ln(\mathbb{P}_\Phi(i))$$

$$\mathbb{P}_\Phi(i) = \frac{|\Phi_i(\mathbf{x}, f, \hat{y}; \lambda)|}{\sum_{j \in [d]} |\Phi_j(\mathbf{x}, f, \hat{y}; \lambda)|}; \quad \mathbb{P}_\Phi = \{\mathbb{P}_\Phi(1), \dots, \mathbb{P}_\Phi(d)\}$$



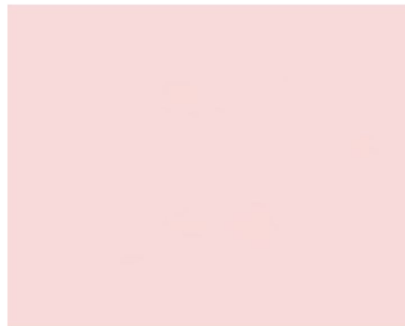
Estimating Explanation Quality - Complexity

- What are some potential issues with using **complexity** as an explainability metric?



Estimating Explanation Quality - Complexity

- **Complexity:** potential issues
 - entropy gives a numerical value, but doesn't tell what's good or bad
 - does not consider spatial structure in explanations (e.g., spatial coherence)
 - low entropy is considered interpretable, but sparse explanations may oversimplify behavior



Example-based Evaluation

- **nDCG**: Normalized Discounted Cumulative Gain
 - popular metric used in information retrieval and ranking tasks to measure ranking quality
 - measures how effectively items are ranked according to their true relevance
 - relevance score (often given by human annotators)

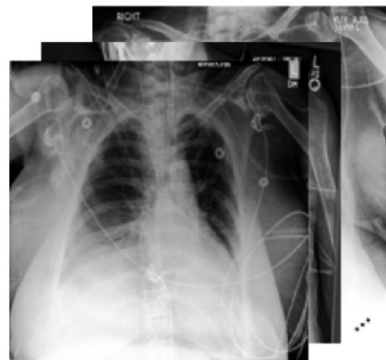
$$\text{nDCG}_p = \frac{DCG_p}{IDCG_p}$$

$$DCG_p = \sum_{i=1}^p \frac{2^{rel_i} - 1}{\log_2(i + 1)}$$

Test Image



Catalogue Images

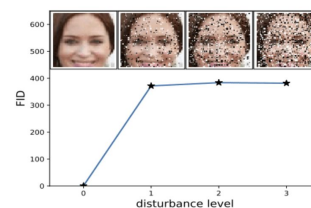
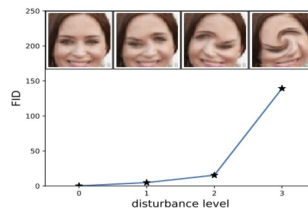
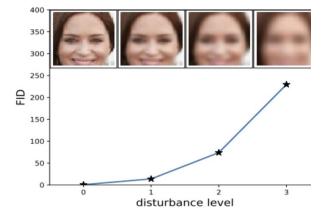
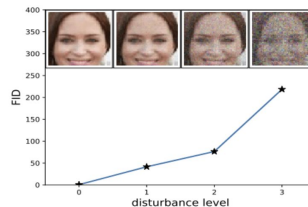


Example-based Evaluation

- **Fréchet Inception Distance**

- evaluate the quality and realism of generated images
- how close the distribution of generated images is to that of real images
- features are extracted using Inception v3 network

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right)$$



Example-based Evaluation

- **Classifier-based evaluation**

- a classifier evaluates if generated counterfactuals genuinely achieve this prediction change

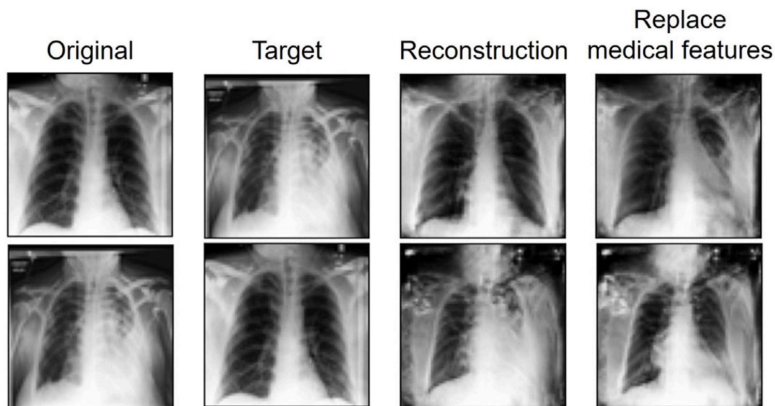
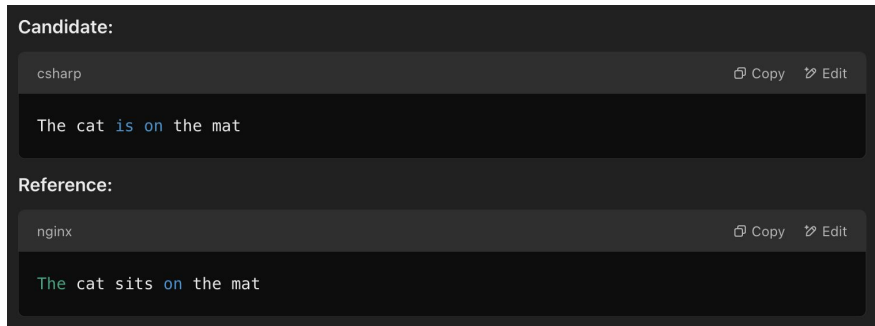


Image Caption Evaluation

- **Bleu score:** measures the quality of machine-generated text
 - compares n-grams (contiguous sequences of n words) between generated and reference
 - higher overlap of n-grams indicates higher quality of generated text
 - relies on exact word matching



The screenshot shows a dark-themed code editor with two sections: 'Candidate:' and 'Reference:'. Each section has a header, a text input field, and a 'Copy' button. The candidate text is 'The cat is on the mat' and the reference text is 'The cat sits on the mat'. The words 'is' and 'on' are highlighted in blue in the candidate text, and 'sits' and 'on' are highlighted in green in the reference text.

```
Candidate:
csharp
The cat is on the mat

Reference:
nginx
The cat sits on the mat
```

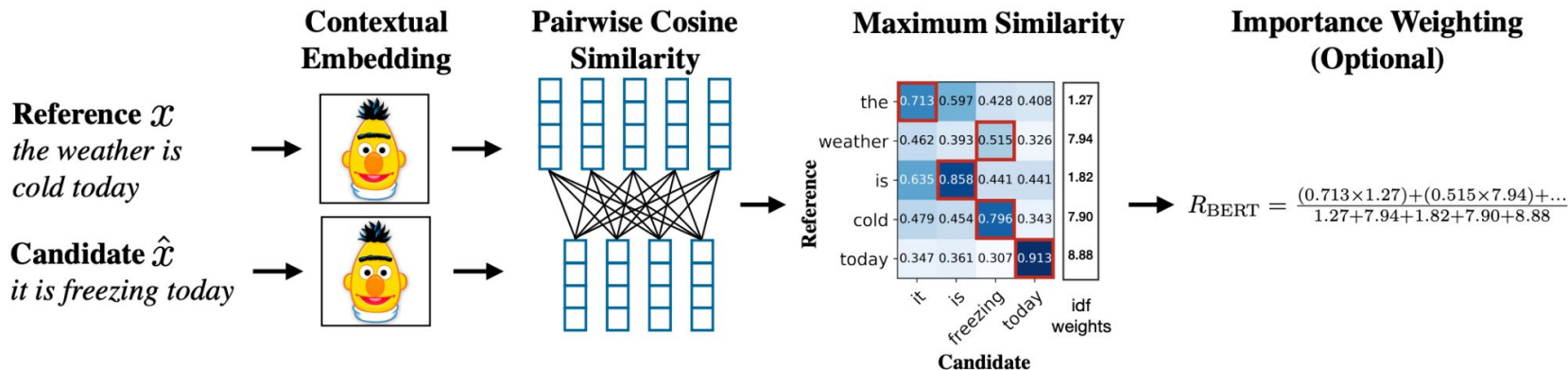
Image Caption Evaluation

- **Rouge-L:** measures the similarity between generated text and reference by finding the longest sequence of words that appear in the same order in both texts
 - longest common subsequence
 - keeps word order but allows for skipped words
 - doesn't capture semantic similarity

Reference (human summary):	"the cat was sleeping on the mat"
Generated (model summary):	"the cat slept on mat"
Longest common subsequence:	"the cat on mat" (length = 4)
ROUGE-L score:	$4/7 \approx 0.57$ (~57%)

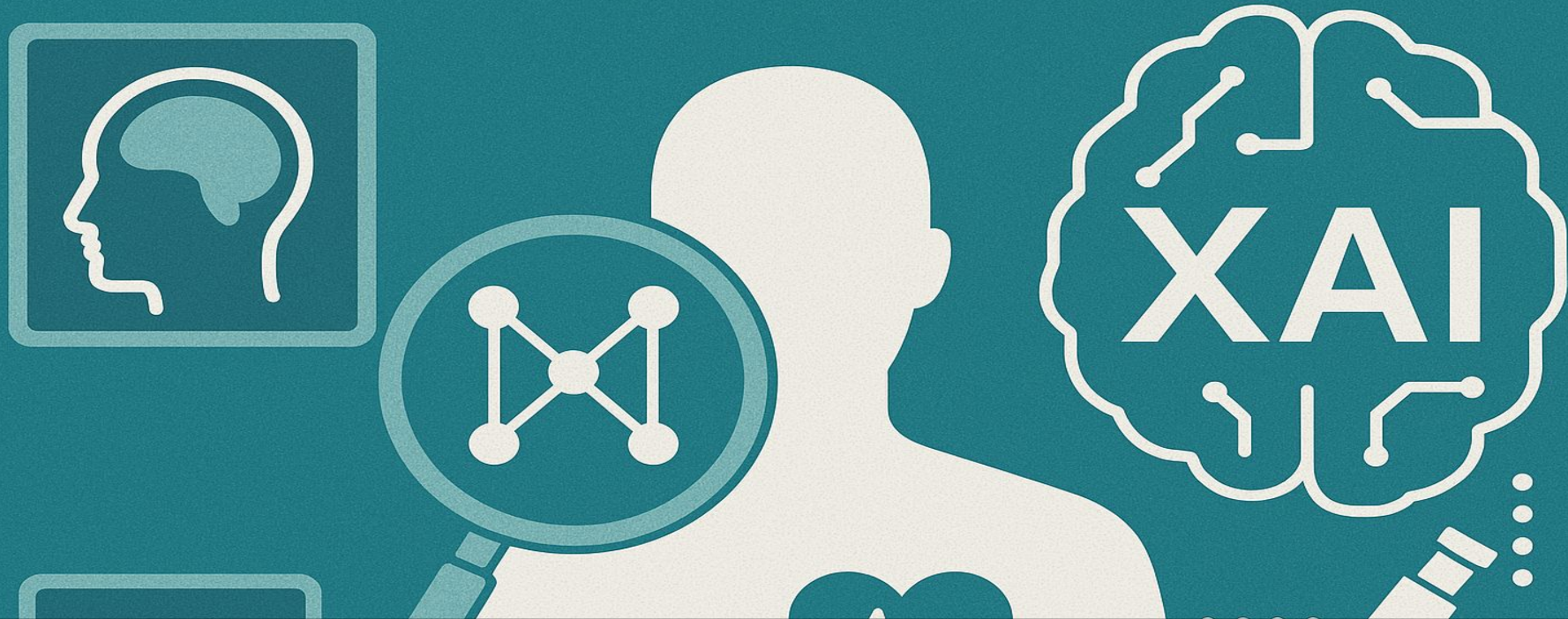
Image Caption Evaluation

- **BERTScore**: semantic similarity metric that evaluates generated captions against reference captions
 - uses pre-trained BERT embeddings to measure meaning similarity rather than word overlap
 - calculates similarity (cosine similarity) scores between embeddings
 - aggregates these scores into a single metric



Hands-on: Explainable AI in a Radiology Scenario

- **Metric to use for saliency maps:**
 - **Localisation:** relevance mass accuracy (Arras et al., 2021)
- **Metric to use for example-based explanations:**
 - **nDCG:** Normalized Discounted Cumulative Gain



INVICTA Spring School: Explainable AI (xAI)

xAI in Healthcare: Fundamentals, Challenges, and the Road to Trustworthy AI

Wilson Silva, Utrecht University